RESEARCH PROTOCOL · EXPERIMENTAL DESIGN · IMPLEMENTATION BLUEPRINT

# Emotion-Contingent Navigation

A preregistered plan for comparing six menu structures across four emotional contexts — and for testing whether switching structure actually helps.

> This document turns the emotional adaptive navigation idea into a testable programme: a pilot, a 6 × 4 mapping study, a causal adaptation study, and an end-to-end system evaluation with an imperfect emotion estimator. It includes the full protocol, measures, analysis plan, sample-size rationale, prototype architecture, ethics, and a paper outline.




**Version 1.0** · 23 October 2026
Prepared for the HCI research project · 2026
Status: planning draft — ethics approval and pilot required before data collection

Built from the four supplied papers, the existing protocol v3.0, and the six-menu brief. Citations marked **VERIFY** could not be confirmed from the folder or public metadata and must not be presented as established evidence.

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### How to use this document

Sections 1–6 define the scientific logic. Sections 7–10 are the three study protocols. Sections 11–13 specify measurement and analysis. Sections 14–17 cover build, ethics, validity, and schedule. Section 18 gives the paper skeleton. The appendices contain task examples, questionnaire items, induction scripts, the event schema, randomisation matrices, and analysis code skeletons. If you have limited time, read **§0, §2, §3, §4, §8, §9, §13**, then use **Track A/B/C** in §17.

# 0 Executive summary

The project asks a simple but surprisingly untested question: **if a user's emotional context changes, should the navigation structure change with it?** Most e-commerce sites keep one menu architecture for everyone. Most adaptive-menu research changes item order, highlights, or recommendations inside a fixed structure. This project compares six genuinely different structures — horizontal bar, vertical sidebar, mega menu, fractal zoom, rounded-square icon grid, and radial lattice — under four experimentally induced emotional contexts, then tests whether switching structure on the basis of that mapping actually improves product finding.

| | | | |
|---|---|---|---|
| **6** menu structures, same taxonomy and products | **4** emotional contexts: happy, frustrated, confused, bored | **24** cells in the core mapping design | **3** linked studies: mapping → causal test → system |

## 0.1 The scientific gap

Four literatures meet here but do not yet connect. Menu research shows that breadth, depth, layout, and visual format change search speed and errors (Miller, 1981; Kiger, 1984; Leuthold et al., 2011; Chiu & Yang, 2024). Adaptive-interface research shows that personalisation can improve satisfaction and engagement, but also that reordering and switching create predictability and learning costs (Findlater & McGrenere, 2004; Gajos et al., 2008; Lavie & Meyer, 2010; Gaspar-Figueiredo et al., 2025). Emotion research shows that affect shapes information processing and search persistence (Russell, 1980; Schwarz & Clore, 1983; Isen, 2001). Navigation research shows that disorientation and frustration are central failure modes (Lazar et al., 2003; Smith, 1996). What is missing is a controlled test that treats **whole menu structure** as the adaptive variable and **emotional context** as the moderator.

The existing protocol (v3.0) proposed four menus and four emotions. This plan extends it in five ways: (1) adds the rounded-square icon grid and the radial lattice, giving six structures that cover linear, panel, recursive-spatial, icon-grid, and radial geometries; (2) formalises the 6 × 4 design, randomisation, and target rotation; (3) replaces optimistic sample sizes with a simulation-based power analysis; (4) adds a causal test that separates *matching* from *mere switching*; and (5) adds an explicit estimator-error simulation so the end-to-end study can answer "how accurate must emotion inference be before adaptation pays off?"

## 0.2 The four emotional contexts

We keep **happiness / positive affect** and **frustration**, and add two states that are specifically diagnostic for navigation: **confusion / lostness** and **boredom / disengagement**. Together they span the two dimensions of the affect circumplex (valence and arousal) plus two states that matter uniquely for wayfinding: uncertainty about *where something is* (confusion) and low motivation to *keep looking* (boredom). Anxiety and curiosity were considered but rejected for the core design; §3.2 explains why.

## 0.3 The six menus

All six render the same four departments, three subcategories per department, six leaf styles per subcategory, and the same 72 fictional watches. The running example is **Sport & Dive → Pro Diver 300m → Titanium Deep**.

| Menu | Interaction | Clicks | What it stresses | Source status |
|---|---|---|---|---|
| **A. Horizontal bar** | Top bar → dropdown → side flyout | 3 | Familiarity; sequential depth; narrow visible set | Chiu & Yang (2024), SSRN preprint VERIFIED |
| **B. Vertical sidebar** | Persistent left tree; accordion expansion | 3 | Persistent overview; vertical scanning; spatial stability | Chiu & Yang (2024), SSRN preprint VERIFIED |
| **C. Mega menu** | One panel shows 3 columns × 6 leaves | 2 | Breadth; all options visible; one fewer click | Protocol v3.0 baseline; standard pattern INTERNAL |
| **D. Fractal tiles** | Recursive 2 × 2 grid; zoom in/out; breadcrumb | 3 | Self-similar spatial memory; recursive depth | Sahraoui et al. (2025) VERIFIED |
| **E. RS icon grid** | Colour-coded rounded-square cards at each level | 3 | Visual salience; icon recognition; grid search | User brief; Chiossi CHI 2026 citation UNVERIFIED |
| **F. Radial lattice** | 4 compass sectors; L2/L3 nodes on outer rings | 3 | Motor efficiency; spatial anchors; radial scanning | Kim et al. (2022) VERIFIED ; Collins CHI 2026 UNVERIFIED |

**What makes this a paper rather than a prototype demo**

The paper does not assume "frustrated users need a sidebar." Study 1 *estimates* the 6 × 4 mapping. Study 2 then freezes that mapping and tests whether using it beats (a) the best static menu and (b) a deliberately mismatched menu. Study 3 tests the complete system with a noisy estimator and a safe transition policy. A null result — for example, one menu wins everywhere, or switching costs cancel the gains — is a publishable finding because it answers the applied question "is emotion-contingent structural adaptation worth it?"

## 0.4 The three studies at a glance

| Study | Question | Design | Output |
|---|---|---|---|
| **Pilot** | Are the menus, tasks, and inductions understandable and tolerable? | N = 8–10; all six menus; all four inductions; think-aloud; timing | Refined protocol, task calibration, variance estimates for power |
| **Study 1 Mapping** | Which menu works best in each emotional context? Is there a menu × context interaction? | N = 72 (full) or 48 (reduced); 6 menus × 4 states; 24 trials; within-subject; Latin-square counterbalancing | 6 × 4 performance/experience matrix; posterior probabilities; frozen policy |
| **Study 2 Causal test** | Does emotion-matched switching beat a static menu and a mismatched menu? | N = 80 (full) or 60 (reduced); 4 conditions × 4 states (static, matched, mismatched, sham); 16 trials; within-subject; known labels | Causal estimate of matching; switching-cost and disruption evidence |
| **Study 3 End-to-end** | Does the system still help when the state estimate is imperfect? | N = 72; static vs hybrid adaptive; 8 trials + free browsing; participant-grouped estimator validation | Estimator accuracy/calibration; end-to-end benefit; break-even accuracy |

## 0.5 Primary outcomes

**Performance:** task success (binary), completion time (log-transformed; censored at timeout), navigation errors, backtracking, category switches, click count, and path efficiency. **Experience:** perceived usability, cognitive load, perceived control, predictability, disorientation, satisfaction, and preference. **Adaptation:** switch acceptance, reversion, perceived disruption, trust, and the cost of the first trial after a switch. **Estimation:** macro-F1, balanced accuracy, per-class recall, confusion matrix, calibration, and decision-curve net benefit.

## 0.6 Feasibility and decisions

### Full programme

N ≈ 234 across three studies; 6–9 months part-time; lab with a 1920 × 1080 desktop; optional eye tracker; ethics approval; one research assistant for counterbalancing and data checks.

### Reduced programme (recommended for a semester)

Pilot + Study 1 (N = 48) + Study 2 (N = 60). Drop Study 3 or run it as a small feasibility study (N = 24) without a deployable estimator. This still yields a complete paper: mapping + causal test + a simulation of estimator error.

### Minimum publishable unit

Pilot + Study 1 with N = 48 and a focused test of four pre-registered state-specific contrasts. This is enough for a CHI Extended Abstract or a thesis chapter, but the causal claim is weaker.

### What the plan deliberately does not do

- It does not use webcam, facial expression, or biometric "emotion detection." Self-report anchors the state; interaction signals are treated as noisy evidence.
- It does not assume that one menu is "best" before the data. Directional predictions are stated, but the mapping is estimated.
- It does not change the menu mid-route. Switches happen at task boundaries, with an explanation and a keep-current option in Study 3.
- It does not hide null results. A global winner, a failed induction, or a switching-cost null are all informative.

#### Three things to confirm before building

**(1) Timeline and sample access.** The full design needs ~234 participants; the reduced design ~118. If you cannot reach that, use Track A in §17 and pre-register a focused, underpowered-by-design study with precision reporting. **(2) Equipment.** Eye tracking is optional; the core plan uses interaction logs and self-report. **(3) Citation status.** The Chiossi et al. (CHI 2026) and Collins et al. (CHI 2026) references could not be verified. Menus E and F are defined operationally here; do not cite unverified claims in the paper.

# 1 Background and related work

This project sits at the intersection of menu design, adaptive user interfaces, and affective computing. The literature is rich in each area, but the three are rarely combined in a single controlled experiment. This section summarises what is established, what the four supplied papers contribute, and where the present study adds new evidence.

## 1.1 Menu structure is not a neutral container

Decades of menu research show that the shape of a hierarchy changes search behaviour. The classic breadth–depth trade-off finds that broader, shallower menus are faster and less error-prone than deep, narrow ones, at least within the ranges tested (Miller, 1981; Kiger, 1984; Snowberry et al., 1983). Two-level menus often outperform deeper hierarchies (Parkinson et al., 1988), which is one reason the mega menu — a panel that exposes several levels at once — can be efficient (Li & Nielsen, 2017). But breadth is not free: a panel with many links increases visual density and the number of distractors, and it can make category boundaries harder to parse.

Orientation matters too. Leuthold et al. (2011) used eye tracking to compare vertical, dynamic, and simple menus and found that vertical menus required less visual effort and produced faster, more successful navigation. Chiu and Yang (2024) extended this to e-commerce menu formats, comparing a horizontal main menu with a vertical local menu and a fully horizontal version. Their eye-tracking data showed that a horizontal main menu plus a vertical local menu improved visual efficiency and satisfaction relative to a dense, multi-category horizontal layout. The practical lesson is not "vertical is always better"; it is that **the fit between orientation, density, and task** matters, and that subjective and physiological measures can diverge.

Visual form also matters. Ahlström et al. (2010) showed that square menus can be faster than traditional hierarchical menus because they exploit the full 2D space and reduce pointing time. Radial menus exploit direction and distance in a different way, and Fitts's law predicts that targets arranged around a central hub can be acquired quickly when the direction is known (Fitts, 1954). The rounded-square icon grid and the radial lattice in this study are not arbitrary novelties; they manipulate **spatial geometry**, **visual salience**, and **motor trajectory**, which are exactly the dimensions that menu research says should matter.

## 1.2 Adaptive menus: benefits, costs, and the switching problem

Adaptive menus change item order, prominence, or availability based on usage. Findlater and McGrenere (2004) compared static, adaptive, and adaptable menus and found that adaptation can help, but user control and awareness moderate the benefit. Gajos et al. (2008) showed that predictability and accuracy trade off: an adaptive interface that moves items can be faster in the moment but slower overall if users cannot predict where things are. Lavie and Meyer (2010) catalogue the benefits and costs of adaptive user interfaces, including the cost of re-learning and the risk of maladaptive changes. This is the core tension for the present project: **a structure that fits the current state may be better, but changing structure destroys spatial memory and imposes a transition cost.**

The supplied RL paper (Gaspar-Figueiredo et al., 2025) is a useful counterpoint. It extends an RL-based adaptive-UI framework with individual human feedback: each participant ranks pairs of adaptation videos, a personalised reward modifier is trained, and a unique RL agent is trained for that participant. In a crossover study with 33 participants across an e-learning and a trip-planning domain, the adaptive interface with human feedback produced higher user satisfaction (estimated −0.97 for non-adaptive, $p < .001$) and higher engagement (−0.23, $p = .016$) than the non-adaptive interface. The paper is important because it shows that **individualised feedback can make adaptation work**, and because it carefully separates adaptive from non-adaptive baselines. But it adapts layout, font size, density, theme, and widget type — not whole menu architectures — and it measures UX rather than product-finding

performance. It also does not test whether the adaptation remains beneficial when the preference model is wrong, nor does it quantify switching costs.

The fractal menu paper (Sahraoui et al., 2025) contributes a different idea: use the entire display as a recursive, self-similar hierarchy. Each menu level is a rectangular area; selecting an area zooms into its children; selecting outside the area zooms out; a breadcrumb supports direct return. The authors engineer the menu with an L-system, validate it in a steel-mill checklist use case, and argue that self-similarity supports proprioceptive memory and orientation in deep hierarchies. Crucially, the paper is a **concept and engineering paper with a use-case validation**, not a controlled comparison. It explicitly says that empirical demonstration remains future work. The present study provides that empirical test, and adds the emotional moderator the fractal paper does not consider.

## 1.3 Emotion, affect, and information seeking

Emotion is not a single variable. The most widely used descriptive model is the valence–arousal circumplex: affective states vary in how pleasant they are and how activated they are (Russell, 1980). Mehrabian and Russell's (1974) PAD model adds dominance, which is relevant to navigation because perceived control is a core part of the experience. Appraisal theories add cognitive structure: emotions arise from evaluations of goal relevance, goal congruence, agency, and certainty (Ortony et al., 1988). This matters because frustration, confusion, and boredom are not just points on a circle; they have different appraisals and different action tendencies.

Affect influences how people process information. Positive affect tends to broaden attention and increase exploratory behaviour (Isen, 2001), which may make a visually richer or more exploratory menu more acceptable. Negative affect narrows attention and increases vigilance, which may make a dense panel feel overwhelming. The affect-as-information account (Schwarz & Clore, 1983) predicts that people may use their current feelings as a cue when judging an interface, so the same menu can be evaluated differently depending on mood. In navigation specifically, disorientation is a well-documented source of frustration (Lazar et al., 2003), and lostness has been operationalised as the mismatch between the user's expected path and the actual path (Smith, 1996). Boredom is the mirror image: low arousal, low interest, and reduced exploration, which can lead to fast guessing or abandonment (O'Brien et al., 2018).

Consumer research adds a domain-specific taxonomy. Richins' (1997) Consumption Emotions Set and Laros and Steenkamp's (2005) hierarchical model distinguish positive and negative consumption emotions, including contentment, excitement, frustration, and disappointment. The present study uses a smaller, experimentally tractable set because repeated measurement of a large taxonomy is impractical. The four states are selected for their **theoretical distinctness**, **relevance to navigation**, and **measurability with brief self-report plus interaction signals**.

## 1.4 The four supplied papers: what they do and do not establish

| Paper | What it contributes | What it does not establish | How this plan uses it |
|---|---|---|---|
| **Chiu & Yang (2024) "Eyes Bring Out Effectiveness?" SSRN 4920862** | Eye-tracking comparison of horizontal vs vertical menu formats in e-commerce; 39 participants; fixation duration, effective focus, revisit rate, SUS, satisfaction; argues for a horizontal main menu + vertical local menu. | One product domain (robot vacuum), a preprint not peer-reviewed, no emotional manipulation, no adaptive policy, no test of structural switching. | Motivates the horizontal bar (A), vertical sidebar (B), and mega menu (C); informs AOI definitions and the optional eye-tracking extension. |
| **Sahraoui et al. (2025) "Fractal Adaptive Menus," EICS Companion '25** | Defines a recursive, self-similar menu; L-system engineering; zoom in/out; breadcrumb; steel-mill checklist use case; argues for proprioceptive memory in deep hierarchies. | No controlled comparison, no performance data against other menus, no emotional context, no user study with naive participants. | Provides the operational definition of menu D (fractal tiles) and the theoretical rationale for spatial memory. |

| | | | |
|---|---|---|---|
| **al. (2025) "Integrating Human Feedback into an RL-Based Framework for AUIs," EASE 2025** | Personalised RL agents trained from pairwise human feedback; 33-participant crossover; QUIS and UES; adaptive > non-adaptive on satisfaction and engagement; releases code and materials. | Adapts layout/density/theme/widgets, not menu architecture; no product-finding task; no emotion; no test of estimator error or switching cost. | Motivates the individualised, human-in-the-loop framing; informs the adaptation policy and the UX measures; provides a baseline logic for comparing adaptive vs static. |
| **Protocol v3.0 "Emotional Adaptive Navigation — Revised Protocol"** | Proposes four menus × four emotions; three studies (mapping, causal test, end-to-end); measurement and ethics plan; selection rule; preregistration checklist. | Four menus only; no RS grid or radial lattice; provisional sample sizes; no simulation-based power analysis; no formal menu-parity controls. | Provides the scientific skeleton. This document extends it to six menus and adds the design, power, and implementation detail needed to run it. |

## 1.5 Theoretical framework and why each menu might fit each state

The study is informed by four mechanisms, each of which makes a different prediction. These are **predictions to test**, not assumptions baked into the design.

| Mechanism | Claim | Design implication |
|---|---|---|
| **Cognitive load / breadth–depth** | Working memory is limited; exposing more options at once reduces depth but increases visual search load. The optimum depends on the user's state and the task. | Mega menu (C) should help when the user wants an overview; deep sequential menus (A, B) should help when the choice set is small and labels are clear. |
| **Wayfinding / spatial memory** | Persistent spatial layouts support cognitive maps; recursive zoom and radial anchors may support proprioceptive memory; reordering destroys it. | Sidebar (B), fractal (D), and radial (F) should reduce lostness for users who can form a spatial map; switching between structures should impose a cost. |
| **Visual salience / affect** | Colour, icons, and motion capture attention; positive affect may increase tolerance for richness, while frustration may make it distracting. | RS grid (E) and radial (F) may increase engagement and reduce boredom, but may increase errors under frustration or confusion. |
| **Motor efficiency / Fitts's law** | Target acquisition time depends on distance and target size; radial and grid layouts can reduce travel and exploit directional gestures. | Radial (F) and grid (E) should be fast when the target location is known; sequential menus (A, B) should be slower but more predictable. |

Combining these mechanisms yields the directional expectations in Table 1.2. They are deliberately **not** the basis for the adaptive policy; the policy is derived from Study 1. They are used to pre-register focused contrasts and to interpret the pattern.

Table 1.2 — Directional expectations (to be tested, not assumed)

| State | Most likely to benefit | Least likely to benefit | Mechanistic reason |
|---|---|---|---|
| **Happiness** | C (mega), E (RS grid), F (radial) | B (sidebar) if it feels plodding | Broadened attention, openness to exploration, tolerance for visual richness. |
| **Frustration** | C (mega), A (horizontal) | D (fractal), F (radial) if motion and spatial demands add load | Narrowed attention, desire for directness and control; fewer clicks and visible options reduce blockage. |
| **Confusion / lostness** | B (sidebar), C (mega), D (fractal with breadcrumb) | E (RS grid) if icons are ambiguous; F if the ring is hard to parse | Need for a stable overview, clear labels, and a visible path back; spatial persistence supports a cognitive map. |
| **Boredom** | E (RS grid), F (radial) | A (horizontal), B (sidebar) as "more of the same" | Under-stimulation; novelty, colour, and motion may restore engagement, at the cost of efficiency. |

### Why the mapping must be empirical

These predictions conflict in interesting ways. For example, under frustration, C is predicted to help because it reduces clicks, but it also presents 18 links at once, which may increase visual load. Under confusion, B is predicted to help because it is persistent, but a persistent tree can also be overwhelming. The study is designed to resolve these trade-offs, not to confirm a slogan.

## 1.6 Contributions

1. **Empirical:** the first 6 × 4 matrix of menu-structure performance and experience across experimentally induced emotional contexts, with uncertainty estimates.
2. **Causal:** a test of whether emotion-matched structural switching beats a static best menu and a mismatched menu, separating matching from mere change.
3. **Methodological:** a simulation-based power analysis for the 6 × 4 design, a menu-parity checklist, and a break-even accuracy analysis for emotion inference.
4. **Applied:** a reusable open-source prototype and event schema for comparing heterogeneous menu architectures over one taxonomy.
5. **Theoretical:** evidence on whether the benefit of adaptation is explained by cognitive load, spatial memory, perceived control, or visual engagement.

# 2 Research questions and hypotheses

The programme asks one overarching question and four specific ones. Each specific question maps onto a study and a pre-registered primary outcome. The hypotheses are directional where theory supports it and exploratory where it does not.

## 2.1 Overarching question

> **Which e-commerce navigation structure best supports product finding in each of four emotional contexts, and does adapting the structure to that empirically derived mapping improve performance and experience compared with a static or mismatched structure?**

## 2.2 Specific questions

| ID | Question | Study | Primary evidence |
|---|---|---|---|
| RQ1 | Do the six menu structures differ in task success, completion time, navigation errors, backtracking, cognitive load, and perceived usability? | Study 1 | Main effects of menu; estimated marginal means |
| RQ2 | Does the relative advantage of the structures change across emotional contexts — i.e., is there a menu × context interaction? | Study 1 | Interaction test; 6 × 4 matrix; focused contrasts |
| RQ3 | Does an emotion-matched adaptive policy outperform a static best-menu policy, a mismatched policy, and a sham-switching control? | Study 2 | Planned contrasts on success, time, errors, load, disorientation |
| RQ4 | Can interaction signals predict self-reported emotional context above a majority-class baseline, and does end-to-end adaptation remain beneficial when the estimate is imperfect? | Study 3 | Participant-grouped validation; macro-F1; calibration; static vs hybrid |
| RQ5 | Are the effects of menu structure mediated by cognitive load, perceived control, and wayfinding/lostness? | Study 1 & 2 | Mediation models; process measures |

## 2.3 Hypotheses

| ID | Hypothesis | Operationalisation | Decision rule |
|---|---|---|---|
| H1 | Menu structure and emotional context interact: the rank order of menus differs across at least some contexts. | Menu × context interaction in the mixed-effects model for log completion time and success. | Likelihood-ratio test $p < .05$ (or Bayes factor > 3) *and* a meaningful spread in at least two state-specific contrasts. |
| H2 | A policy that selects the Study 1 best-supported menu will improve task performance relative to the best static menu. | Matched vs static contrast in Study 2. | Superiority on time/errors with non-inferiority on success (margin 5 percentage points). |
| H3 | Emotion-matched adaptation will outperform emotion-mismatched adaptation. | Matched vs mismatched contrast. | Superiority on the primary composite; direction consistent across at least three of four states. |

| | | | |
|---|---|---|---|
| **H5** | Interaction signals will predict self-reported state above the majority-class baseline, but with meaningful uncertainty. | Participant-grouped cross-validation; macro-F1 and balanced accuracy. | Macro-F1 > baseline + .10 with 95% CI excluding baseline; report confusion matrix and calibration. |
| **H6** | End-to-end benefits will be smaller than the known-label (oracle) benefit and will depend on estimator confidence and transition stability. | Study 3 vs Study 2; confidence-stratified analysis. | Hybrid > static on at least one primary outcome, but hybrid < matched (known labels); benefit increases with confidence. |
| **H7** | Switching imposes a measurable cost: the first trial after a structure change will be slower and more error-prone than a stable trial. | Switch trial vs stable trial within Study 2/3. | Switch cost > 0 on log time or errors, with 95% CI excluding zero. |
| **H8** | Cognitive load, perceived control, and lostness will mediate the effect of menu structure on performance. | Mediation with bootstrapped indirect effects. | Indirect effect 95% CI excludes zero; report proportion mediated. |

## 2.4 Primary and secondary outcomes

### Primary (confirmatory)

- **Task success** — correct product detail page within 120 s.
- **Completion time** — log-transformed; failures censored at 120 s.
- **Navigation errors** — wrong branch, wrong product, invalid transition.
- **Cognitive load** — NASA-TLX after each block.
- **Perceived usability** — UMUX-Lite after each block; SUS at the end.
- **Disorientation / control** — 4-item scale after each block.

### Secondary (exploratory / mechanism)

- Backtracking, category switches, clicks, path efficiency.
- Menu open/close count; hover/focus dwell; search attempts (if enabled).
- Perceived predictability, trust, satisfaction, preference ranking.
- Switch acceptance, reversion, perceived disruption.
- Eye tracking (optional): time to first fixation, fixation count, scanpath, AOI transitions.
- Individual differences: domain expertise, online shopping frequency, need for cognition.

## 2.5 What would count as failure — and why it is still informative

| Observed pattern | Interpretation |
|---|---|
| **No menu × context interaction; one menu wins everywhere** | Emotion-contingent structural adaptation is unnecessary in this domain. The applied recommendation is a stable global menu; the theoretical claim is that menu structure is robust to mild emotional context. |
| **Interaction present, but matched adaptation does not beat static** | Switching costs, task variation, or insufficient policy precision offset the cell-level gains. This is a boundary condition, not a failed study. |
| **Matched beats static, but mismatched also beats static** | The benefit is change/novelty/attention, not matching. The sham control and the direction of the mismatch effect distinguish these. |
| **Matched works with known labels but not with the estimator** | The bottleneck is inference or transition policy. The break-even accuracy analysis says how much better the estimator must be. |
| **Users frequently revert or report disorientation** | User control and structural stability may outweigh performance. The design recommendation is to keep the current layout and offer adaptation as an option, not a default. |

**Pre-registration rule**

The primary outcome, the 6 × 4 interaction test, the four state-specific focused contrasts, the Study 2 contrasts, and the policy-selection rule must be frozen before confirmatory data collection. Exploratory analyses are labelled as such. The mapping is frozen after Study 1 and must not be revised after looking at Study 2 outcomes.

# 3 The four emotional contexts

Emotion is treated as an experimental context, not as a clinical state and not as something that can be read directly from clicks. The four contexts are chosen because they are distinct, relevant to navigation, and measurable with brief self-report plus behavioural evidence. The system estimates **self-reported context**, not "true emotion."

## 3.1 The four states and why they fit

| State | Operational definition | Induction (mild, reversible) | Behavioural signals (noisy) | Why it matters for menus |
|---|---|---|---|---|
| **Happiness / positive affect** | Participant reports positive valence and a good mood after a brief positive experience; moderate arousal. | 90-s amusing video clip + a positive "you found a great deal" scenario; or a success-feedback task. Pilot to select the stronger induction. | More direct choices, fewer reversals, longer exploration, higher tolerance for visual richness. | Tests whether positive affect broadens exploration and makes richer menus (E, F, C) more acceptable. |
| **Frustration** | Participant reports irritation or frustration after a mild, controlled goal blockage; high arousal, negative valence. | 90-s "buggy search" task: valid queries return "No results," the filter resets, and a neutral message appears. Capped at three attempts; no personal failure feedback. | Repeated clicks, rapid reversals, repeated menu open/close, pauses after failure, higher click rate. | Tests whether frustration narrows attention and favours direct, low-click, high-visibility structures (C, A). |
| **Confusion / lostness** | Participant reports uncertainty about where a product belongs or which path to choose; negative/neutral valence, moderate arousal. | 90-s ambiguous categorisation task: place a product into one of four overlapping categories with no definitions or feedback. | Long pauses, category switching, backtracking, repeated menu inspection, inconsistent paths. | Tests whether confusion favours a stable overview and a visible path back (B, C, D with breadcrumb). |
| **Boredom / disengagement** | Participant reports low interest or under-stimulation after a repetitive, low-challenge task; low arousal, negative/neutral valence. | 90-s repetitive catalogue task: verify the price of 30 identical watch straps, one by one, with no variation. | Long idle gaps, unusually fast guesses, reduced exploration, lower engagement ratings, early exits. | Tests whether boredom favours novel, colourful, or spatially engaging menus (E, F) over familiar lists (A, B). |

## 3.2 Why these two extra states — and why not anxiety or curiosity

The brief asked for two additional emotions beyond happiness and frustration. The recommendation is **confusion/lostness** and **boredom/disengagement**. The rationale is that they are the two states most directly implicated in navigation failure and in the engagement literature, and they occupy different quadrants of the affect space:

- **Confusion** is the cognitive-affective signature of wayfinding failure. It is the state that menu structure is most obviously supposed to fix, and it is measurable through backtracking, category switching, and self-report.
- **Boredom** is the low-arousal, low-engagement state that a visually rich or spatially novel menu might repair. It is the natural counterpart to confusion: one is "I don't know where it is," the other is "I don't care enough to keep looking."
- Together with happiness (positive, moderate arousal) and frustration (negative, high arousal), they cover the valence × arousal space and add a motivational dimension that a pure circumplex misses.

Table 3.2 — Alternatives considered and why they are not in the core design

| Alternative | Why it is attractive | Why it is not a core state |
|---|---|---|
| **Anxiety** | High-arousal negative state; relevant to purchase decisions and risk. | Overlaps with frustration; harder to induce ethically and to distinguish behaviourally; better treated as a moderator or a future extension. |
| **Curiosity / interest** | Positive, high-arousal; directly relevant to exploratory browsing. | Overlaps with happiness and is difficult to induce independently of the task; could be a fifth state in a follow-up. |
| **Sadness** | Low-arousal negative; distinct from boredom in appraisal. | Less tied to menu structure; inducing sadness in a shopping study raises ethical and demand-characteristic concerns. |
| **Delight** | Positive, high-arousal; relevant to aesthetic menus. | Transient and hard to measure reliably; more of an evaluation reaction than a stable context. |
| **Cognitive load** | Central to menu performance. | Not an emotion; treated as a mediator and measured with NASA-TLX. |
| **Stress** | Physiological, measurable. | Broad and non-specific; ethical and equipment burden; weak mapping to menu geometry. |

## 3.3 Measurement model: self-report anchors, behavioural evidence, no mind-reading

Each state is measured in three ways. First, **self-report** is the primary label: a 9-point SAM valence and arousal rating (Bradley & Lang, 1994) plus four discrete 7-point items (happy, frustrated, confused, bored). Second, **behavioural signals** are logged continuously and used as noisy features for the estimator. Third, **manipulation checks** verify that the induction produced the intended state. The system never claims to detect emotion from behaviour alone; it predicts the self-reported context under the tested conditions.

State labels are derived with a pre-registered rule. The intended state is the label if (a) the corresponding discrete item is the highest of the four and (b) it is rated ≥ 5 on the 7-point scale. If the highest item is below 5 or two items are tied, the trial is labelled **mixed/neutral** and analysed separately. This avoids forcing every participant into a category and makes induction failure visible.

## 3.4 Induction protocol

All inductions are mild, short, and reversible. They are embedded before a block of six navigation trials, not during a trial. A neutral reset (60-s nature video or a simple filler task) separates blocks. The order of blocks is counterbalanced with a Latin square. The full scripts are in Appendix C.

| State | Cover story | Materials | Manipulation check | Stop rule / debrief |
|---|---|---|---|---|
| **Happiness** | "Before the next set, we'll show a short clip from a comedy show used in mood research." | 90-s validated amusing clip; headphones; volume fixed; no brand content. | SAM valence ≥ 6; happy item ≥ 5; arousal 4–7. | If distress > 6/10, stop; debrief that the clip was for mood, not evaluation. |
| **Frustration** | "The site's search is being tested and may be unreliable; please try up to three times." | Buggy search: valid query returns "No results," filter resets; neutral error copy; 90 s cap. | Frustration item ≥ 5; SAM valence ≤ 4; arousal ≥ 5. | Stop after 3 attempts; immediate debrief; no mention of participant error; offer break. |
| **Confusion** | "We're checking how people categorise a new product; there is no right answer." | Four overlapping categories, no definitions, no feedback; 90 s. | Confusion item ≥ 5; SAM valence ≤ 5; arousal 4–6. | Reveal the ambiguity; explain the purpose; offer break. |
| **Boredom** | "Please verify the prices of these identical straps for the catalogue." | 30 identical items; repetitive clicking; no variation; 90 s. | Boredom item ≥ 5; SAM arousal ≤ 4; valence ≤ 5. | Stop at 90 s; debrief; offer a stimulating break. |

**Pilot criteria for a usable induction**

An induction is retained only if, in the pilot (N = 8–10): (1) at least 60% of participants label the intended state as dominant; (2) mean intensity is ≥ 5/7; (3) fewer than 20% report a different state as dominant; (4) no participant reports distress above 6/10; (5) the induction takes ≤ 2.5 min including the check. If an induction fails, revise the script or replace it before the main study; do not proceed with a weak manipulation.

## 3.5 Carry-over, reactivity, and demand characteristics

Emotional states can carry over between blocks, and repeated self-report can make participants aware of the hypothesis. Three controls address this. First, a neutral reset and a 2-min break separate blocks. Second, the purpose of the study is described broadly as "how people find products in different website layouts," not as "we are testing whether emotion changes menu choice." Third, the post-experiment suspicion check asks what participants thought the study was about; participants who correctly guess the adaptation hypothesis are retained but the analysis is repeated without them as a sensitivity check.

Order effects are handled by counterbalancing the four emotion blocks across participants using a Latin square (four sequences: ABCD, BCDA, CDAB, DABC). Within each block, the six menus are presented in a second Latin square (six positions). The combination yields 24 sequences; participants are assigned randomly with stratification by gender and prior online-shopping frequency. Trial position is included as a covariate in the analysis.

**What this study does not claim**

It does not diagnose emotion, measure clinical anxiety or depression, or infer emotion from facial expression. It does not claim that the induced states are identical to real shopping emotions. It tests whether *mild, experimentally induced, self-reported contexts* moderate the relative performance of menu structures. That is a narrower and more defensible claim.

# 4 The six menu structures

The independent variable is **menu structure**, not a cosmetic theme. All six structures expose the same information architecture and the same products, but they differ in geometry, visible breadth, click depth, spatial persistence, and motor demands. This section defines each structure precisely enough to build it, then specifies the controls that keep the comparison fair.

## 4.1 Shared information architecture

The store is a fictional watch retailer with four departments, three subcategories per department, and six leaf styles per subcategory — 72 products in total. Every menu renders this same tree; only the presentation changes. The running example throughout is **Sport & Dive → Pro Diver 300m → Titanium Deep**.

Table 4.1 — Taxonomy used by all six menus

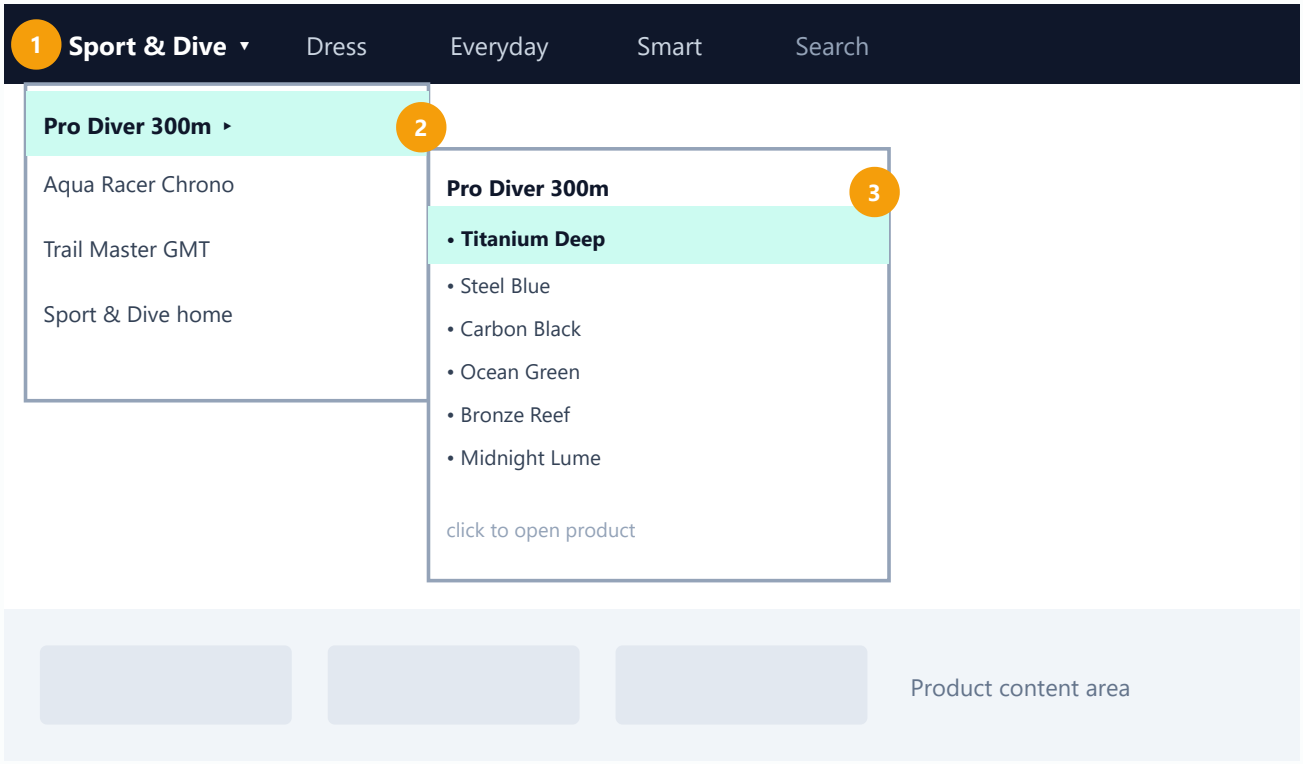
| Department (L1) | Subcategory (L2) | Leaf styles (L3) — six per subcategory |
|---|---|---|
| **Sport & Dive** | Pro Diver 300m | Titanium Deep · Steel Blue · Carbon Black · Ocean Green · Bronze Reef · Midnight Lume |
| | Aqua Racer Chrono | Regatta White · Gulf Blue · Carbon Track · Sunset Orange · Steel Grey · Night Dash |
| | Trail Master GMT | Alpine Sand · Forest Green · Granite Grey · Summit Blue · Ember Red · Midnight Trail |
| **Dress & Formal** | Heritage Classic | Ivory Dial · Rose Gold · Anthracite · Moon Silver · Bordeaux · Onyx |
| | Midnight Dress | Deep Navy · Black Tie · Sapphire · Graphite · Pearl White · Champagne |
| | Artisan Moonphase | Celestial Blue · Warm Silver · Slate Grey · Aurora · Copper Brown · Starlight |
| **Everyday & Casual** | Field Standard | Khaki Canvas · Olive · Stone · Sandstone · Charcoal · Navy Weave |
| | Urban Minimal | Matte White · Concrete Grey · Ink Black · Soft Beige · Steel Mesh · Deep Teal |
| | Weekender | Sunset Clay · Ocean Stripe · Cream · Moss · Denim · Rust |
| **Smart & Hybrid** | Smart One | Graphite · Silver · Rose · Midnight · Titanium · Forest |
| | Hybrid Active | Black Sport · Blue Sport · Green Sport · Orange Sport · White Sport · Grey Sport |
| | Connected Classic | Heritage Black · Heritage Brown · Steel Blue · Cream · Slate · Burgundy |

Product attributes for task descriptions: case size (38/40/42/44 mm), strap (steel, titanium, leather, rubber, mesh, fabric), movement (automatic, quartz, solar, hybrid), water resistance (50/100/200/300 m), dial colour, and price band. Each leaf style has a unique combination; target descriptions use two or three attributes rather than the exact product name to avoid label-matching.

## 4.2 Operational definitions and wireframes

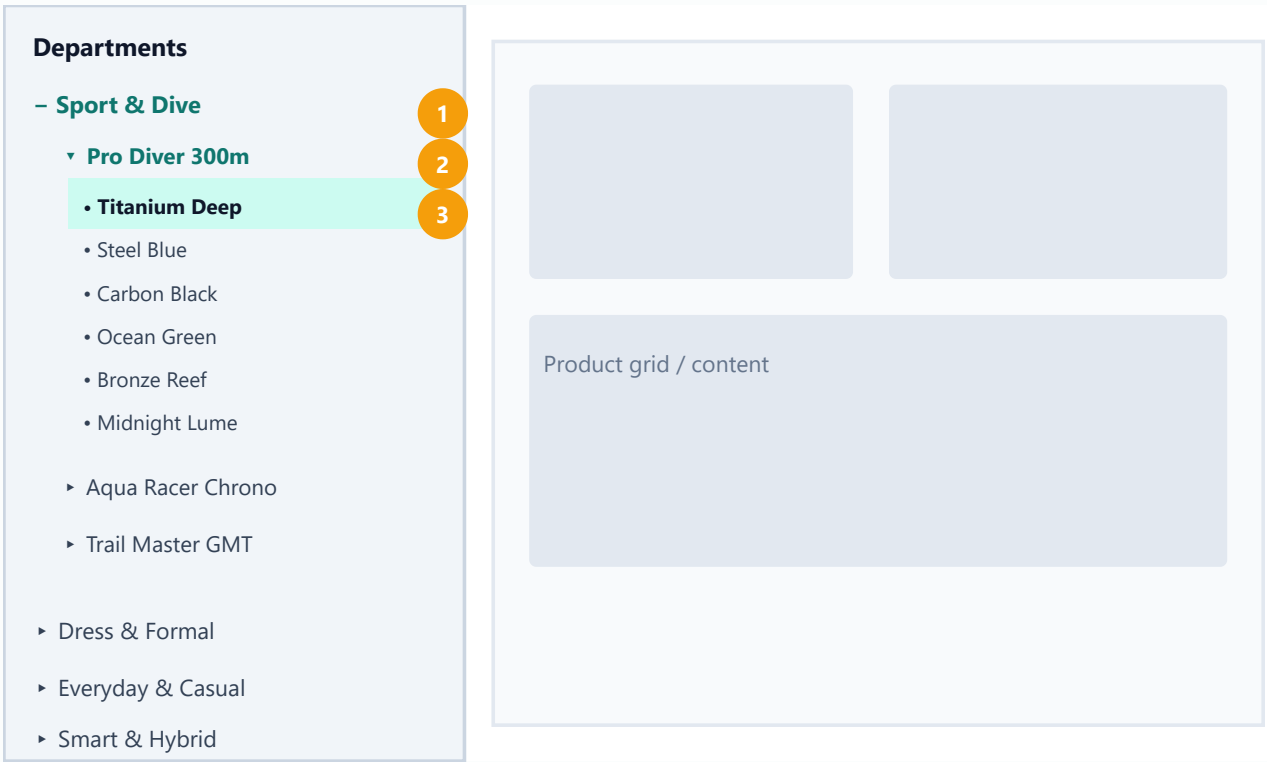
The wireframes below are **interaction specifications**, not final visual designs. They show what is visible at each step, where the click targets are, and how the user moves between levels. The build must preserve these structural properties; the visual skin can be refined in the pilot as long as parity controls (§4.3) are respected.

**A · Horizontal bar — top bar, dropdown L2, side flyout L3**




Click 1 opens the L2 dropdown; click 2 opens the L3 flyout; click 3 selects the leaf. Three clicks. Only one L2 and one L3 set are visible at a time. The flyout closes on outside click.

**B · Vertical sidebar — persistent left tree, nested accordion**




The tree persists on the left. Click 1 expands the department, click 2 expands the subcategory, click 3 selects the leaf. Three clicks. The main content area is always visible, so the tree can be used as an overview.

**C · Mega menu — one panel, 3 columns, all L3 visible**

1 Sport & Dive ▾ | Dress | Everyday | Smart

| Pro Diver 300m | Aqua Racer Chrono | Trail Master GMT |
|---|---|---|
| **Titanium Deep** 2 | Regatta White | Alpine Sand |
| Steel Blue | Gulf Blue | Forest Green |
| Carbon Black | Carbon Track | Granite Grey |
| Ocean Green | Sunset Orange | Summit Blue |
| Bronze Reef | Steel Grey | Ember Red |
| Midnight Lume | Night Dash | Midnight Trail |

All 3 subcategories and 18 leaf styles are visible at once.

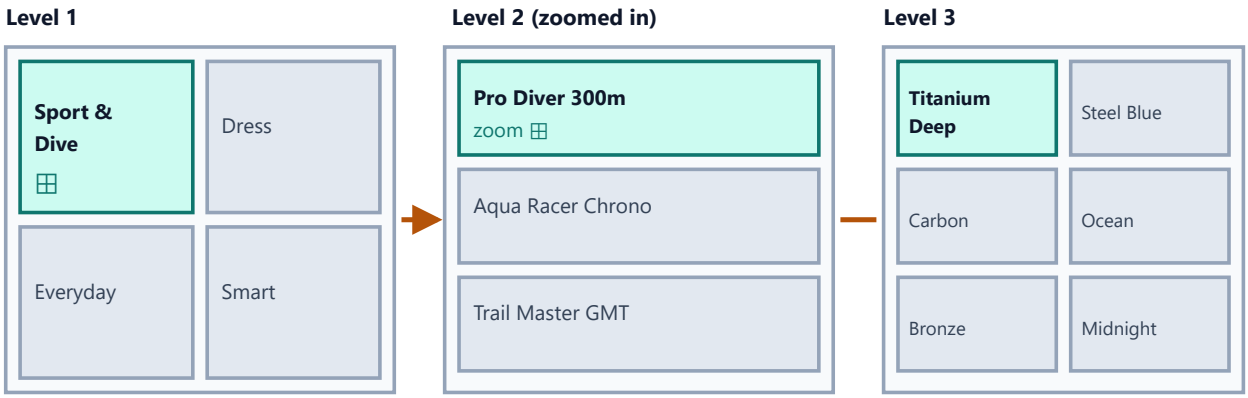
**2 clicks to leaf (one fewer than A/B/D/E/F)**

Product content area

Click 1 opens the panel showing all three subcategories and all 18 leaf styles. Click 2 selects the leaf. Two clicks, but high visual density. This is the Protocol v3.0 baseline.

**D · Fractal / recursive tile menu — zoom into the parent tile**

Home > Sport & Dive > Pro Diver 300m

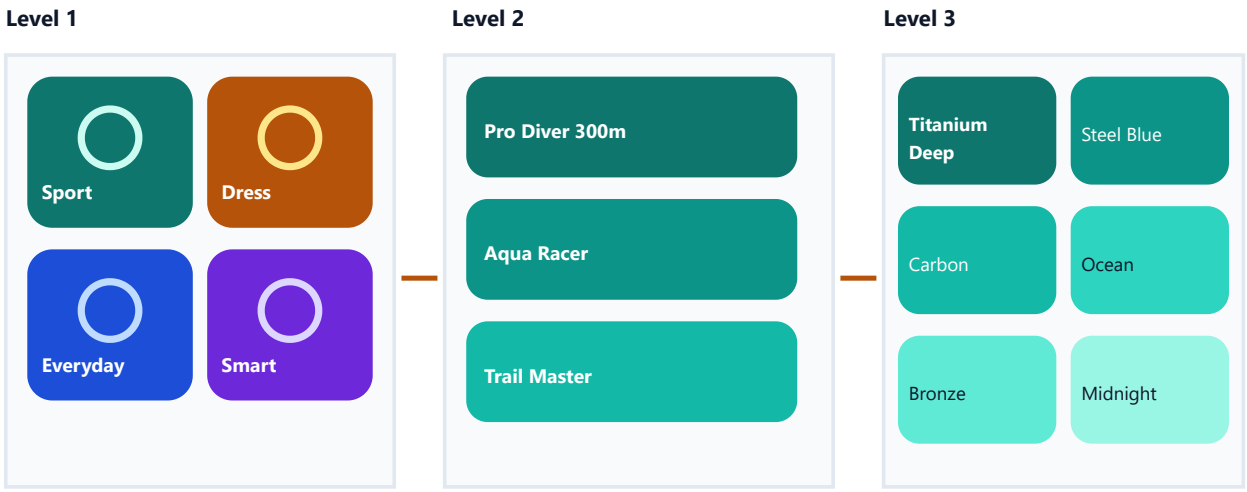



Click a tile to zoom in; click outside the tile or use the breadcrumb to zoom out. The same rectangular layout recurs at every level.

**3 clicks to leaf · recursive spatial layout · breadcrumb + zoom-out affordance**

Each parent tile occupies a rectangular area; selecting it zooms the viewport so its children fill the same geometric layout. Three clicks. Spatial self-similarity is the hypothesised memory advantage.

**E · Rounded-square (RS) visual icon-grid — colour-coded cards at each level**

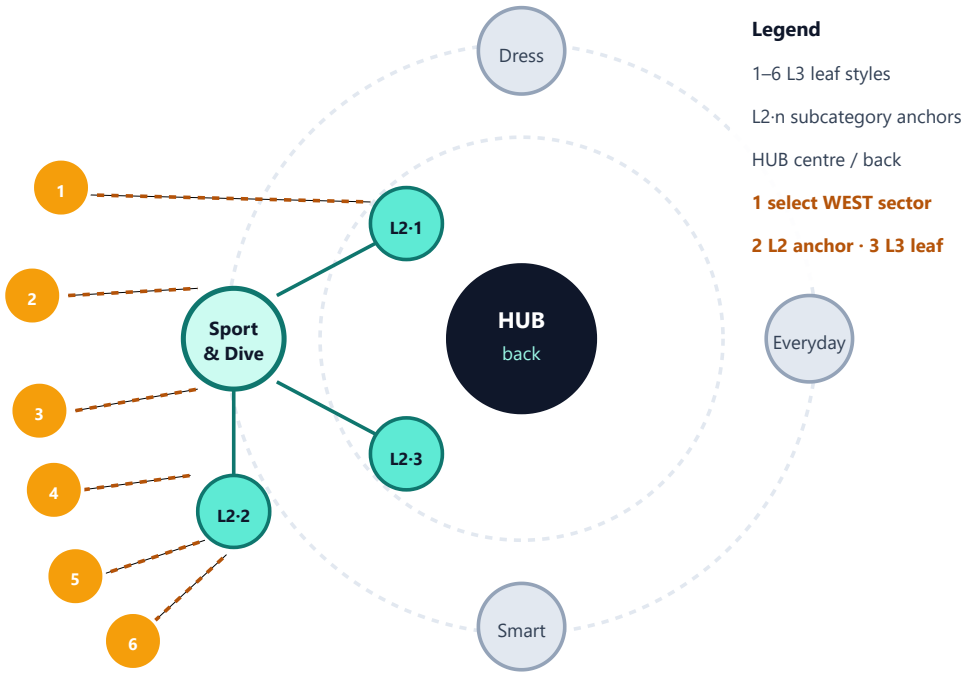



Each level is a grid of large rounded-square cards with an icon, a label, and a category colour; type is not the only cue.

**3 clicks · 4 → 3 → 6 cards · high visual salience · colour and icon redundancy**

Four large cards, then three subcategory cards, then six product cards. Three clicks. The design stresses icon recognition and colour-coded spatial grouping; it is expected to be engaging but potentially less precise under confusion.

**F · Radial / circular lattice — compass sectors, hub, outer rings**




Departments sit at compass sectors; selecting one projects L2 anchors onto an inner ring and L3 leaves onto the outer ring.

Four fixed sectors around a central hub. Selecting WEST (Sport & Dive) projects three L2 anchors and then six L3 leaves onto the outer ring. Three clicks. The design trades list scanning for directional pointing and spatial anchoring. Adapted from Kim et al. (2022), who used a lattice of visual anchors for gaze; here the input is mouse/trackpad.

## 4.3 Parity and confound control

The structures are intentionally different, so some confounds are intrinsic. The goal is not to make them identical — that would destroy the manipulation — but to ensure that every difference is either part of the construct or measured and modelled. Table 4.2 is the build contract.

**Table 4.2 — What is held constant, what varies, and how it is handled**

| Property | Control | Why / analysis consequence |
|---|---|---|
| **Taxonomy, labels, products** | One JSON source of truth; all six menus read the same tree. Identical department, subcategory, and leaf labels. | Ensures the information is the same. A label-parity unit test fails the build if any menu has a missing or renamed node. |
| **Product content** | Same product cards, prices, images, attributes, and detail pages after selection. | Outcome differences cannot be attributed to product information. |
| **Viewport and device** | 1920 × 1080, same browser build, same mouse, same zoom level, full-screen mode. | Removes screen-size and input-device variance. Log actual viewport and device. |
| **Typography and palette** | Same font family, sizes, spacing scale, and 4-colour department palette across all menus. E uses larger colour areas; the palette itself is shared. | Prevents E from having a unique colour advantage. Colour-to-department mapping is counterbalanced across participants. |
| **Icons** | Same icon set for each department and subcategory; icons are small in A–D/F and large in E. | Icon meaning is constant; icon *prominence* is part of E's manipulation. |
| **Animation** | 180–220 ms ease-in-out for all transitions; respect `prefers-reduced-motion` ; no decorative motion. | Prevents D/F from gaining a novelty advantage through animation. Motion type differs (zoom/rotate vs expand/fade), so perceived motion is a manipulation check. |
| **Hit targets** | Minimum 44 × 44 px for every interactive element; 8 px minimum spacing. | Avoids a trivial motor advantage. F's ring nodes must meet the same minimum. |
| **Click depth** | A/B/D/E/F = 3 clicks; C = 2 clicks. | Click savings is a genuine property of the mega menu, not a bug. Clicks are logged and tested as a mediator. A matched-depth sensitivity analysis excludes C or adds a confirmation step in a follow-up. |
| **Visible breadth** | A/B/D/E/F show 4 → 3 → 6 options; C shows 4 → 18 options. | Breadth is part of the construct. Log the number of visible options at each step and include it as a covariate/mediator. |
| **Persistence** | B keeps the tree visible; A/C are transient; D/F replace the viewport; E replaces the grid. | Persistence is part of the construct (wayfinding). Measure perceived persistence and use it in mediation. |
| **Search** | Search is visibly present but disabled with a neutral "search is unavailable in this session" message. Participants are asked to use the menu. | Forces menu use. A follow-up field study should allow search; in this experiment, search attempts are logged as a signal of menu failure. |
| **Breadcrumb** | Present in D and F; present in A/B/C only after a department is selected; E shows a compact breadcrumb. | Breadcrumbs are part of the wayfinding affordance. Their presence is logged; a follow-up can isolate breadcrumb effects. |
| **Back behaviour** | In-page "back" control plus browser back; both logged. No destructive navigation. | Prevents a menu from being penalised for a missing escape route. |
| **Loading** | No artificial delays; all content preloaded; local server; 60 fps target. | Prevents network variance from masquerading as a menu effect. |

**The click-depth problem — state it, do not hide it**

The mega menu (C) reaches a leaf in two clicks; the other five need three. That is a real advantage of mega menus and a plausible reason frustrated users like them. But it means a C-vs-A difference cannot be attributed to "layout" alone. The analysis therefore reports (a) the raw structural comparison, (b) a mediation analysis with clicks as the mediator, and (c) a matched-depth sensitivity analysis on A/B/D/E/F only. If the C advantage disappears when clicks are equated, the paper says so. This is more honest than adding a fake third click to C and pretending the menus are equal.

## 4.4 Why these six? Coverage of the design space

| Menu | Geometry | Breadth at choice | Persistence | Motor demand | Unique contribution |
|---|---|---|---|---|---|
| **A · Horizontal** | Linear, top | Low (3) | Transient | Horizontal + vertical | The familiar e-commerce baseline; tests whether convention beats fit. |
| **B · Sidebar** | Linear, left | Low–medium (3–9) | High | Vertical | Persistent overview; tests wayfinding and spatial stability. |
| **C · Mega** | 2D panel | High (18) | Transient | 2D grid | Maximum breadth, minimum clicks; tests the breadth–depth trade-off. |
| **D · Fractal** | Recursive 2D | Low–medium (3–6) | High (spatial) | 2D zoom | Self-similar spatial memory; tests recursive wayfinding. |
| **E · RS grid** | 2D icon grid | Low–medium (3–6) | Medium | 2D pointing | Visual salience and icon recognition; tests engagement vs precision. |
| **F · Radial** | Polar | Low–medium (3–6) | Medium | Directional | Fitts's-law efficiency and spatial anchors; tests a non-Cartesian layout. |

## 4.5 Source status

| Menu | Source in the brief | Verification status and how it is used |
|---|---|---|
| A, B | Paper P3: Chiu & Yang (2024), SSRN 4920862 | **VERIFIED** The paper is a 2024 SSRN preprint by Tseng-Ping Chiu and Ya-Chun Yang. It compares horizontal and vertical menu formats with eye tracking. Cite it as a preprint, not as a peer-reviewed journal article. |
| C | Protocol v3.0, page 5 baseline | **INTERNAL** The mega menu is the protocol's baseline, not an external paper. Cite the protocol as an internal planning document if needed; otherwise describe C as a standard mega-menu pattern (Li & Nielsen, 2017). |
| D | Sahraoui et al. (2025), ACM EICS Companion | **VERIFIED** "Fractal Adaptive Menus: Concept, Engineering, and Use Case." The paper is a concept/engineering paper with a use case; it does not report a controlled comparison. The plan supplies that test. |
| E | Chiossi et al. (2026), ACM CHI | **UNVERIFIED** No matching paper was found in the folder or in public metadata searches. The design is therefore specified operationally in this plan. Do not attribute claims to Chiossi et al. until the PDF and DOI are confirmed. If the citation is real, add it before submission; if not, describe E as a design pattern inspired by square/icon menus (Ahlström et al., 2010). |
| F | Kim et al. (2022), CHI; Collins et al. (2026), CHI | **VERIFIED** Kim, Ham, Ahn, and Lee (2022), "Lattice Menu: A Low-Error Gaze-Based Marking Menu Utilizing Target-Assisted Gaze Gestures on a Lattice of Visual Anchors," CHI '22, DOI 10.1145/3491102.3501977. **UNVERIFIED** The Collins et al. (2026) citation could not be confirmed; do not rely on it. F is adapted from the Kim et al. lattice from gaze to mouse. |

### Academic-integrity note

Two of the six menu attributions in the brief could not be verified. The plan keeps the designs because they are well-defined and testable, but it does **not** claim that Chiossi et al. or Collins et al. reported the effects described. Before submitting the paper, either obtain the PDFs and verify the claims or remove the attributions. The reference list marks both as "unverified — do not cite as evidence."

# 5 Platform, product taxonomy, and task bank

The experiment needs a controlled storefront, a task generator, and an event log precise enough to reconstruct every path. This section specifies the product set, the task design, the success criteria, the logging schema, and the build-versus-reuse choices. The goal is a platform that a second researcher can reproduce from a repository and a seed.

## 5.1 The fictional store

The store is called **Meridian Timepieces** (fictional; no real brand, logo, or trademark). It sells 72 watches across four departments. Product images are either generated 3D renders or licensed stock images with brands removed; all prices are fictional. The visual design is deliberately clean and consistent across conditions: white background, slate text, teal accent, four department colours used as redundant cues. The same product detail page is used after a correct selection in every menu, so the post-selection experience cannot explain differences.

| Attribute | Values used in the task bank |
|---|---|
| **Department** | Sport & Dive; Dress & Formal; Everyday & Casual; Smart & Hybrid |
| **Subcategory** | Three per department (see Table 4.1) |
| **Leaf style** | Six per subcategory; unique name and image |
| **Case size** | 38, 40, 42, 44 mm |
| **Strap** | Steel, titanium, leather, rubber, mesh, fabric |
| **Movement** | Automatic, quartz, solar, hybrid |
| **Water resistance** | 50, 100, 200, 300 m |
| **Dial colour** | Blue, black, green, white, grey, brown, silver, champagne, orange |
| **Price band** | £180–£240; £240–£420; £420–£900; £900+ |

Each leaf style is assigned a unique attribute vector. A build-time test checks that every target description maps to exactly one product; if two products share the same description, the task is rejected and regenerated. This prevents ambiguous “find the blue watch” tasks from contaminating the design.

## 5.2 Task design

The primary task is **known-item product finding**: the participant is given a short, natural-language shopping goal and must reach the correct product detail page using the menu. The task never names the exact product; it describes two or three attributes. This forces category navigation rather than string matching, while keeping the target unambiguous.

> **Example task (the running example)**
>
> **“You’re looking for a titanium dive watch with 300 m water resistance and a deep blue dial. Find it.”**
> Correct path: **Sport & Dive → Pro Diver 300m → Steel Blue** (or the leaf whose attribute vector matches; the build-time uniqueness test decides). The exact label “Titanium Deep” is not shown in the task, so the participant must use the hierarchy and the product attributes.

**Difficulty** is manipulated as a within-participant covariate, not as a fully crossed factor. Three levels are defined by three orthogonal properties: (1) serial position of the department and subcategory in the menu, (2) semantic distance between the task wording and the leaf label, and (3) number of plausible distractor leaves with the same colour or material. Each participant sees 24 targets, balanced 8 easy / 8 medium / 8 hard. The assignment of

difficulty to the 24 menu × state cells is rotated across participants so difficulty cannot be confounded with a condition.

Table 5.1 — Twelve example tasks (the full bank contains 72 targets plus matched variants)

| ID | Task wording (attribute description) | Correct department | Correct subcategory | Correct leaf | Diff. |
|---|---|---|---|---|---|
| **T01** | Titanium dive watch, 300 m, deep blue dial | Sport & Dive | Pro Diver 300m | Steel Blue | Easy |
| **T02** | Lightweight titanium dive watch, 300 m, black dial | Sport & Dive | Pro Diver 300m | Titanium Deep | Medium |
| **T03** | 42 mm automatic trail watch, green dial, fabric strap | Sport & Dive | Trail Master GMT | Forest Green | Medium |
| **T04** | Rose-gold dress watch, ivory dial, leather strap | Dress & Formal | Heritage Classic | Ivory Dial | Easy |
| **T05** | Black-tie watch, deep navy dial, 40 mm | Dress & Formal | Midnight Dress | Deep Navy | Medium |
| **T06** | Moonphase complication, celestial blue dial | Dress & Formal | Artisan Moonphase | Celestial Blue | Hard |
| **T07** | 38 mm field watch, khaki canvas strap, automatic | Everyday & Casual | Field Standard | Khaki Canvas | Easy |
| **T08** | Minimalist watch, matte white dial, steel mesh strap | Everyday & Casual | Urban Minimal | Matte White | Medium |
| **T09** | Weekend watch, sunset clay dial, rubber strap | Everyday & Casual | Weekender | Sunset Clay | Hard |
| **T10** | Graphite smartwatch, titanium case, 44 mm | Smart & Hybrid | Smart One | Graphite | Medium |
| **T11** | Hybrid sport watch, orange accent, rubber strap | Smart & Hybrid | Hybrid Active | Orange Sport | Hard |
| **T12** | Connected classic, burgundy leather strap, 42 mm | Smart & Hybrid | Connected Classic | Burgundy | Hard |

## 5.3 Success, errors, and path efficiency

| Outcome | Definition and coding rule |
|---|---|
| **Success** | 1 if the correct product detail page (matching product ID) is reached within 120 s; otherwise 0. Automated coding from the event log; no experimenter judgement. |
| **Completion time** | Time from task onset (instruction disappears) to the correct product page render. Log-transformed. Failures are right-censored at 120 s and analysed with a survival model as a sensitivity analysis. |
| **Wrong department** | First department selected is not the target department. Counted once per trial. |
| **Wrong subcategory** | Correct department, wrong subcategory selected before success. Counted per trial. |
| **Wrong leaf** | Any leaf selection other than the target before success. Counted per trial. |
| **Backtrack** | A transition from a deeper level to a shallower level without reaching a leaf. Counted per trial. |
| **Category switch** | Selecting a different department or subcategory after a previous selection. Counted per trial. |
| **Path efficiency** | Minimum required clicks / actual clicks. Minimum is 2 for C and 3 for A/B/D/E/F. Values > 1 are capped at 1; values ≤ 0 are set to missing. |
| **First-click correctness** | 1 if the first department click is the target department. A process measure of category knowledge. |
| **Search attempts** | Number of clicks on the disabled search control. A signal of menu failure; not a success path. |
| **Timeout / give-up** | No correct product within 120 s. The participant can press “I can’t find it” to end early; both are failures, but the reason is logged. |

## 5.4 Event logging and data schema

Every interaction is logged as a JSON line with a client timestamp (`performance.now()`) and a server receipt timestamp. The client timestamp is used for within-trial timing; the server timestamp is used for data integrity. Events are batched every 500 ms and flushed on page unload; an offline queue prevents data loss if the connection drops.

| Field | Type | Description |
|---|---|---|
| **event_id** | UUID | Unique event identifier; idempotent ingestion. |
| **participant_id** | string | Pseudonymous ID; no name, email, or IP in the research dataset. |
| **session_id** | UUID | One per session; links events to consent and questionnaires. |
| **study** | enum | `pilot`, `s1`, `s2`, `s3`. |
| **block_id / trial_id** | string | Emotion block and trial identifiers. |
| **task_id / target_id** | string | Task bank ID and correct product ID. |
| **menu_id** | enum | `A` – `F`. |
| **state_label** | enum | `happy`, `frustrated`, `confused`, `bored`, `mixed`. |
| **condition** | enum | `static`, `matched`, `mismatched`, `sham`, `hybrid` (study-specific). |
| **event_type** | enum | See the list below. |
| **t_client / t_server** | float / ISO | Millisecond client time; server receipt time. |
| **elapsed_ms** | int | Milliseconds since trial start. |
| **level** | int | 0 = root, 1 = department, 2 = subcategory, 3 = leaf. |
| **element_id / label** | string | Stable ID and visible label of the clicked/hovered element. |
| **parent_id** | string | Parent node ID, for reconstructing the path. |
| **x / y** | int | Viewport coordinates of the pointer event. |
| **input_type** | enum | `mouse`, `trackpad`, `keyboard`, `touch`. |
| **meta** | JSON | Event-specific fields (e.g., `visible_options`, `switch_reason`, `confidence`). |

**Event types:** `session_start`, `consent`, `demographics`, `practice_start`, `block_start`, `induction_start`, `induction_end`, `checkin`, `trial_start`, `menu_open`, `menu_close`, `node_hover`, `node_click`, `leaf_select`, `product_view`, `back`, `search_attempt`, `timeout`, `give_up`, `trial_end`, `block_end`, `adaptation_shown`, `adaptation_accepted`, `adaptation_reverted`, `questionnaire`, `session_end`.

```
// Example event (truncated)
{
  "event_id": "evt_01HZX ... ",
  "participant_id": "p_0042",
  "session_id": "ses_01HZX ... ",
  "study": "s1",
  "block_id": "block_frustrated",
  "trial_id": "trial_07",
  "task_id": "T02",
  "target_id": "prod_titanium_deep",
  "menu_id": "C",
  "state_label": "frustrated",
  "event_type": "node_click",
  "t_client": 18432.7,
  "elapsed_ms": 3920,
  "level": 2,
  "element_id": "sport.prodiver",
  "element_label": "Pro Diver 300m",
  "parent_id": "sport",
  "x": 412, "y": 268,
  "input_type": "mouse",
  "meta": { "visible_options": 18, "menu_open_ms": 740 }
}
```

## 5.5 Platform architecture

The recommended build is a single-page React + TypeScript application with a small Node/PostgreSQL backend. The front end owns the menu renderers, the experiment timeline, and the event buffer. The back end owns randomisation, condition assignment, data ingestion, and exports. The analysis pipeline reads a versioned CSV/JSONL export and a frozen analysis script.

| Layer | Recommended choice | Why |
|---|---|---|
| **Front end** | React 18 + TypeScript + Vite; Zustand for session state; CSS modules or Tailwind. | Component reuse across six menus; strong typing for the taxonomy; fast local development. |
| **Menu rendering** | Shared `MenuRenderer` contract; A/B/C with Radix/Headless UI primitives; D/F with D3 + SVG; E with CSS Grid + Framer Motion. | Reuse accessible primitives where possible; custom geometry where the structure demands it. |
| **Experiment runner** | Custom timeline inspired by jsPsych; `performance.now()` for timing; server-side sequence assignment. | jsPsych is excellent for standard stimuli, but the six custom menus need a React-native runner. Borrow its timeline and data model. |
| **Back end** | Node + Express/Fastify + PostgreSQL (or Supabase for a managed option). | Reliable ingestion, simple exports, role-based access, easy local deployment. |
| **Data store** | Normalised tables for participants, sessions, trials, events, questionnaires; JSONB for event metadata. | Fast queries for analysis; schema validation at ingestion; raw JSONL retained for reproducibility. |
| **Deployment** | Lab server on a wired network; optional cloud deployment behind authentication. | Avoids internet latency and third-party data processing; supports offline fallback. |
| **Analysis** | R 4.x: `lme4`, `lmerTest`, `emmeans`, `brms`, `survival`, `pROC`; Python for simulation if preferred. | Mixed models, Bayesian policy estimation, survival analysis, and calibration are all mature in R. |

## 5.6 Build versus reuse

There is no single library that provides all six menus over one taxonomy. Reuse the primitives and build the geometry. The table lists candidate sources; check licence, maintenance, and accessibility before adopting.

| Need | Candidate libraries / repositories | Recommendation |
|---|---|---|
| **A/B/C primitives** | Radix UI NavigationMenu; Headless UI Menu; shadcn/ui NavigationMenu; MUI Menu; Ant Design Menu. | Use Radix or Headless UI for focus management and ARIA; restyle to the shared palette. Do not adopt a full design system just for the menu. |
| **C · mega menu** | `jasonrundell/react-mega-menu` (small, MIT-style; verify); many CSS-only patterns. | Use as a structural reference; the panel must show 3 columns × 6 leaves and meet the parity contract. A custom component is likely simpler than adapting a library. |
| **D · fractal** | `jeanvdd/Fractal` ("Fractal Adaptive Menu"; verify licence and completeness); D3 `zoomable treemap` / `d3-zoom` + `d3-hierarchy`. | Use D3 for the zoom/transition; the paper's L-system is a conceptual reference, not a production dependency. Build the 4 → 3 → 6 grid directly. |
| **F · radial** | `react-planet`; `react-pie-menu`; `react-radial-menu`; D3 `arc` and `radialLine`. | Use D3 for precise angles and hit targets; generic radial menus often assume a single ring and cannot project L2/L3 onto concentric rings. |
| **E · icon grid** | CSS Grid + Framer Motion; Radix primitives; icon set (Lucide, Phosphor). | Build directly. The hard part is the product-card content and the colour system, not the grid. |
| **Experiment runner** | jsPsych 7; lab.js; PsychoJS. | Borrow the timeline/data model; do not force the custom menus into a stimulus plugin unless the timing tests pass. |
| **Eye tracking (optional)** | Tobii Pro Lab / Tobii SDK; PyGaze; iMotions. | Use Tobii Pro Lab for AOI definition and export; pre-register the fixation filter and AOIs. |
| **Stats** | R `lme4`, `brms`, `emmeans`, `survival`, `pROC`, `rmcorr`; Python `statsmodels`, `PyMC`. | R for the primary mixed models; Python only if the team is more fluent there. Freeze the analysis script before data lock. |

## 5.7 Accessibility and standardisation

All menus should meet WCAG 2.2 AA where feasible: keyboard operability, visible focus, 4.5:1 text contrast, 3:1 non-text contrast, reduced-motion support, and semantic roles. However, the confirmatory experiment standardises on mouse/trackpad to compare motor performance; keyboard input is supported and logged, and participants who prefer it are not excluded. Input modality is included as a covariate, and a separate accessibility session can be run if the sample includes keyboard-only users. Do not claim full accessibility compliance without an audit.

**Build acceptance criteria (gate to pilot)**

- Every one of the 72 products is reachable in every menu, with the same number of semantic levels.
- Every target description maps to exactly one product (uniqueness test).
- All six menus show the same labels and product data from one JSON source.
- All interactive targets are ≥ 44 × 44 px at 1920 × 1080; no overlap or clipping.
- Transitions are 180–220 ms and do not drop below 50 fps on the lab machine.
- Event logs reconstruct the full path for 100% of test trials; no missing `trial_end`.
- Randomisation is reproducible from a seed and audited server-side.
- No console errors; no external network requests during a trial.

# 6 Study design overview

The programme is sequential by design. Study 1 estimates the mapping; Study 2 freezes it and tests causality; Study 3 tests the complete system with a noisy estimator. This avoids the circularity of deriving a policy and testing it on the same data.

## 6.1 Why three studies rather than one

| Design | What it can show | What it cannot show |
|---|---|---|
| **Single 6 × 4 experiment** | Whether menus differ across induced states; a descriptive mapping. | Whether using the mapping causally helps; whether switching costs erase the gain; whether a noisy estimator works. |
| **Single adaptive-vs-static experiment** | Whether an adaptive system beats a static one. | Whether the benefit comes from matching, from change/novelty, or from a better global menu. |
| **Three-study sequence** | Mapping, causal matching, and end-to-end feasibility, each with a clean counterfactual. | Long-term learning and real-world adoption; those are future work. |

## 6.2 Design summary

| Study | N (full) | N (reduced) | Design | Trials / session | Primary outcomes |
|---|---|---|---|---|---|
| **Pilot** | 10 | 8 | All menus, all inductions; think-aloud; timing. | 24 + practice; ~90 min | Task success, SUS, manipulation check, technical errors. |
| **Study 1** | 72 | 48 | 6 menus × 4 states within-subject; 24 trials; Latin-square block/menu order; target rotation. | 24 + 6 practice; ~90 min | Success, log time, errors, NASA-TLX, UMUX-Lite, disorientation. |
| **Study 2** | 80 | 60 | 4 conditions (static, matched, mismatched, sham) × 4 states; 16 trials; within-subject; known labels. | 16 + 4 practice; ~70 min | Success, log time, errors, switching cost, perceived disruption, load, trust. |
| **Study 3** | 72 | 24–48 | Static vs hybrid adaptive; 8 trials + 5-min free browsing; estimator validation. | 8 + free browse; ~70 min | Macro-F1, calibration, success/time, switch acceptance, trust. |

The "full" column is the recommended confirmatory sample. The "reduced" column is the minimum for a semester project; it is powered for larger effects and must be reported with precision, not as a definitive null. §13 gives the power curves and the assumptions behind them.

## 6.3 Randomisation and counterbalancing

Study 1 has two crossed order factors: four emotion blocks and six menus within each block. A full randomisation of 24 cells would confound order with the cell-specific targets at small N. Instead, use two Latin squares:

- **Emotion block order:** four sequences — ABCD, BCDA, CDAB, DABC — assigned evenly across participants. Each state appears in each serial position exactly once across the four sequences.
- **Menu order within block:** a 6 × 6 balanced Latin square, so each menu appears in each within-block position once. A different row is used for each block to avoid the same menu always following the same state.

- **Target rotation:** six target sets of 24 tasks, each balanced by department and difficulty. Across participants, each target set is paired with each block order and menu order equally often. The full matrix is in Appendix E.
- **Practice:** one practice trial per menu (not analysed) before the main blocks, to separate menu learning from the experimental effect. Practice targets are not reused in the main trials.
- **Breaks:** a 2-min neutral reset between blocks; participants can take a longer break without losing compensation.

Study 2 uses the same block-order logic but four conditions per state. The four conditions are presented in a 4 × 4 Latin square within each block; across participants, each condition appears in each position equally often. Study 3 has two conditions (static, hybrid) and uses an ABBA/BAAB crossover to balance order and carry-over.

## 6.4 Session length and burden

| Segment | Study 1 | Study 2 | Study 3 | Notes |
|---|---|---|---|---|
| **Consent, demographics, baseline affect** | 12 min | 10 min | 10 min | Includes colour-vision check and audio check. |
| **Instructions and practice** | 12 min | 10 min | 8 min | 6 practice trials (one per menu) in Study 1/2. |
| **Inductions + checks** | 16 min | 16 min | 12 min | 4 blocks; 2.5 min induction + 1 min check each. |
| **Main trials** | 20 min | 12 min | 12 min + 5 min browse | 24 trials × ~45 s; 16 trials × ~45 s; 8 trials + free browsing. |
| **Questionnaires** | 18 min | 12 min | 15 min | SUS per menu; NASA-TLX/UMUX-Lite per block; adaptation items. |
| **Debrief and suspicion check** | 8 min | 6 min | 8 min | Includes neutral-mood check and resources. |
| **Total** | **~86 min** | **~66 min** | **~70 min** | Add a 10-min buffer; pilot confirms. |

## 6.5 Recruitment, inclusion, and compensation

Recruit adults aged 18–65 who are fluent in English and have normal or corrected-to-normal vision. No prior participation in the pilot or in another study of the programme. Colour vision is screened with a short Ishihara-style test; the palette is designed to be distinguishable for the most common deficiencies, so colour-vision deficiency is not an automatic exclusion — it is recorded and analysed as a covariate. Participants must be able to use a mouse or trackpad; keyboard-only users can be accommodated in a separate session.

Use a university participant pool and a reputable online panel (e.g., Prolific) for the reduced design. Compensate at £12–£15 per hour or course credit with a comparable alternative. Do not use performance-contingent bonuses: they would confound emotion induction and risk pressure. Target a 20% attrition/technical-failure buffer; replace participants who fail the pre-registered data-quality checks, not those whose results are inconvenient.

### Stopping and data-quality rules

Collect in waves (e.g., 4 × 18 participants in Study 1) and check only **data quality** between waves: missing logs, failed randomisation, session length, and technical errors. Do not check condition effects or stop early for significance. Pre-register the final N. Exclude a session only if (a) the event log is incomplete for more than 10% of trials, (b) the participant did not complete at least 75% of trials, (c) the participant reports a technical failure that prevented menu use, or (d) the participant was previously in the pilot. Report all exclusions and the reasons in a flow diagram.

# 7 Pilot study

The pilot is not a miniature version of Study 1. It is a **measurement and feasibility study** whose job is to make the main study interpretable: Are the menus usable? Do the inductions work? Are the tasks too easy or too hard? Is the session tolerable? Are the logs complete? The pilot data are not included in the confirmatory analysis.

## 7.1 Goals and success criteria

| Goal | What is measured | Criterion to proceed |
|---|---|---|
| **Menu usability** | Task success, time, errors, SUS, think-aloud comments, observed confusion. | Every menu has ≥ 80% success on practice tasks; no menu is unusable without a design fix; all targets reachable. |
| **Induction validity** | Dominant-state check, intensity, credibility, distress rating. | ≥ 60% intended dominant; mean intensity ≥ 5/7; no distress > 6/10; induction ≤ 2.5 min. |
| **Task difficulty** | Success, time, first-click correctness, verbal reports. | Overall success 75–90%; easy > medium > hard in the expected direction; no floor/ceiling. |
| **Timing and burden** | Session length, trial length, fatigue ratings, breaks requested. | Session ≤ 100 min; trial median ≤ 45 s; fatigue not extreme. |
| **Instrument quality** | Completion time, missingness, straight-lining, comprehension probes. | < 5% missing; no item misunderstood by more than one participant. |
| **Technical reliability** | Missing events, clock drift, rendering errors, frame rate. | 100% of practice trials reconstructable; no unhandled errors; ≥ 50 fps. |

## 7.2 Design and sample

N = 8–10 participants (the reduced track can use 8). A within-subject walkthrough: all six menus, all four inductions, six practice trials, and a subset of main trials (e.g., 12 rather than 24) to test timing. Participants think aloud during the first pass and retrospectively after each block. The pilot uses the same consent, questionnaires, and debrief as Study 1, with additional open-ended probes.

## 7.3 Procedure

| Time | Activity | Notes |
|---|---|---|
| **0–10** | Consent, demographics, baseline affect, colour vision | Same forms as Study 1; add a "prior experience with menu types" item. |
| **10–20** | Demo and think-aloud instructions | Explain that we are testing the website, not the participant; encourage candid comments. |
| **20–35** | Menu walkthrough (A–F) | One practice task per menu; experimenter observes and notes confusion; no help unless the participant is stuck for 60 s. |
| **35–50** | Induction block 1 + check + 3 trials | Counterbalance the induction order across pilot participants. |
| **50–65** | Induction block 2 + check + 3 trials | Include a neutral reset and a fatigue rating. |
| **65–80** | Induction block 3 + check + 3 trials | Probe whether the state feels credible and whether the task feels like a real shopping task. |
| **80–95** | Induction block 4 + check + 3 trials | Stop early if distress; debrief immediately. |
| **95–110** | Questionnaires and retrospective interview | SUS per menu, NASA-TLX, open-ended: what was confusing, what felt fast, what felt frustrating. |

## 7.4 What the pilot changes

The pilot can change the **implementation** and the **measurement**, but not the confirmatory hypotheses. Permitted changes: fix a menu bug; adjust hit-target size; clarify instructions; replace a task that is ambiguous; adjust the induction script; drop an unusable questionnaire item; recalibrate the session length. Changes that would alter the scientific question (e.g., dropping a menu, changing the four states after seeing results) require a protocol amendment and must be reported.

**Pilot outputs**

- A revised task bank with difficulty calibration (success and time by task).
- Variance components for the power simulation: participant intercept SD, residual SD, target SD, menu×state interaction SD.
- A manipulation-check report for each induction (dominant-state %, mean intensity, adverse events).
- A technical QA report (missing events, frame rate, browser console errors).
- A frozen protocol and analysis script for pre-registration.

# 8 Study 1 — the 6 × 4 mapping

Study 1 is the empirical heart of the programme. It estimates how six menu structures perform and feel under four induced emotional contexts, and it produces the mapping that Study 2 tests causally. It is a within-subject experiment with 24 trials per participant, four emotion blocks, and six menus per block.

## 8.1 Objective and pre-registered predictions

**Objective:** estimate the 6 × 4 matrix of performance and experience; test whether the relative ranking of menus depends on emotional context (H1); and derive a policy with explicit uncertainty. The directional expectations in Table 1.2 are used to pre-register four focused contrasts, but the omnibus interaction is the primary gate.

| Focused contrast | Prediction | Rationale |
|---|---|---|
| **Frustration: C vs A** | C (mega) faster and fewer errors than A (horizontal) | Fewer clicks and all options visible; directness under goal blockage. |
| **Confusion: B vs E** | B (sidebar) lower disorientation than E (icon grid) | Persistent tree and explicit labels support a cognitive map; icons may be ambiguous. |
| **Boredom: E vs A** | E higher engagement and faster than A | Visual salience and novelty restore arousal; A is familiar but dull. |
| **Happiness: C vs B** | C faster than B | Broadened exploration and tolerance for density; fewer clicks. |

## 8.2 Design

| Factor | Levels | Type | Notes |
|---|---|---|---|
| **Menu structure** | A, B, C, D, E, F | Within-subject | Six menus per emotion block; order balanced by a 6 × 6 Latin square. |
| **Emotional context** | Happy, frustrated, confused, bored | Within-subject | Four blocks; order balanced by a 4 × 4 Latin square; neutral reset between blocks. |
| **Task difficulty** | Easy, medium, hard | Within-subject covariate | 8 targets per level per participant; rotated across cells. |
| **Trial position** | 1–24 | Covariate | Included to absorb fatigue/learning; not a primary factor. |
| **Target set** | 6 sets × 24 tasks | Between-participant rotation | Balances specific products across cells; target random intercept in the model. |

**Trials per participant:** 24 main trials (6 menus × 4 states), plus 6 practice trials (one per menu). Each trial has a 120-s cap; median expected 25–45 s. The session is ~90 min including consent, inductions, questionnaires, and debrief.

## 8.3 Participants and sample size

**Target N = 72** for the full confirmatory study; **N = 48** for the reduced semester version. The full sample provides ~75% power to detect a small-to-medium omnibus interaction (f = 0.10) and > 90% power for f = 0.15 under the assumptions in §13. The reduced sample is powered for f ≈ 0.15 and for the four focused contrasts if their effects are medium. The pilot updates the variance components before the final N is fixed.

**Inclusion:** 18–65, fluent English, normal/corrected vision, able to use a mouse/trackpad, no prior participation in the pilot. **Exclusion (pre-registered):** incomplete logs for > 10% of trials; < 75% of trials completed; technical failure

preventing menu use; prior participation. Colour-vision deficiency is recorded, not automatically excluded. Report a CONSORT-style flow diagram.

## 8.4 Materials

- The six-menu prototype (§5), running locally at 1920 × 1080, full-screen, with search disabled.
- 72 product pages, 72 task descriptions, six balanced target sets, and six practice tasks.
- Four induction scripts and media: comedy clip (happy), buggy search (frustrated), ambiguous categorisation (confused), repetitive price check (bored).
- Neutral reset video (60 s nature footage) and a 2-min break timer.
- Questionnaires: SAM valence/arousal, four discrete emotion items, NASA-TLX (raw), UMUX-Lite after each block, SUS after each menu, disorientation/control scale, demographics, suspicion check.
- Headphones, consent forms, debrief sheet, and a stop-rule card for the experimenter.

## 8.5 Procedure

| Min | Step | Detail |
|---|---|---|
| **0–12** | Welcome, consent, screening | Information sheet, consent, demographics, baseline SAM and discrete emotions, colour-vision check, audio check. Emphasise voluntary participation and the right to stop. |
| **12–24** | Instructions and practice | Cover story: "We are comparing website layouts for finding watches." Explain the task, the 120-s limit, the disabled search, and the menu-specific interaction. One practice trial per menu (not analysed). Confirm understanding with a short quiz. |
| **24–36** | Block 1: induction + check + 6 trials | 2.5-min induction, 1-min manipulation check, 6 trials in a balanced menu order, 1-min post-block check, 2-min neutral reset. If the state has faded (intensity < 4), re-induce briefly before the next trial. |
| **36–48** | Block 2 | Same structure with the next state in the Latin-square order. |
| **48–60** | Block 3 | Same; offer a longer break if requested. |
| **60–72** | Block 4 | Same; final post-block check. |
| **72–88** | Questionnaires | SUS for each menu (order counterbalanced), NASA-TLX summary, disorientation/control, adaptation expectations, demographics, suspicion check. |
| **88–95** | Debrief | Explain the inductions, the purpose, and the hypotheses; answer questions; provide support resources; thank and compensate. |

## 8.6 Trial procedure (exact sequence)

1. **Task screen (5 s or until "Start"):** the shopping goal is displayed in neutral language. The participant clicks "Start" when ready.
2. **Navigation:** the menu appears with the store home page. The participant uses the menu to reach the product. Search is visibly disabled. The 120-s timer starts at "Start."
3. **Selection:** clicking a leaf opens the product detail page. The system checks the product ID. If correct, the trial ends with a neutral "Response recorded" message. If incorrect, the participant can continue; wrong leaves are logged but do not end the trial.
4. **Timeout / give-up:** at 120 s, or if the participant presses "I can't find it," the trial ends with "Time is up. Moving on." No performance feedback, praise, or criticism.
5. **Inter-trial interval:** 3 s blank screen with a fixation cross, then the next task.
6. **Check-in:** after trials 3 and 6 of each block, a 2-item micro-check (valence and arousal) appears; the full discrete-emotion check appears at the end of the block. This limits reactivity while providing trial-level anchors.

**Why neutral feedback?** Praise or error messages would induce additional emotion and contaminate the block. The product page itself is the success signal. The experimenter does not comment on performance during the session.

## 8.7 Emotion block procedure

| Step | What happens | Why |
|---|---|---|
| **Baseline** | SAM + four discrete items before the induction. | Establishes within-person change; detects carry-over. |
| **Induction** | 2.5-min scripted task or clip, standardised across participants. | Creates the context without using the navigation task itself. |
| **Manipulation check** | SAM + discrete items; dominant-state rule applied. | Verifies the intended state; provides the label for analysis. |
| **Six trials** | Menus in the balanced order; targets from the assigned set. | Estimates the 6 × 1 slice of the matrix for this state. |
| **Post-block check** | SAM + discrete items + NASA-TLX + UMUX-Lite. | Measures fade, load, and usability; supports dose-response analysis. |
| **Reset** | 60-s neutral video + 2-min break; offer water. | Reduces carry-over between states. |

**If the induction fails:** the trial block is retained in the intention-to-treat analysis (randomised state), and a complier-average causal effect (CACE) analysis uses only blocks where the intended state was dominant. This avoids dropping data based on a post-treatment variable. The failure rate is reported by state and induction order.

## 8.8 Outcomes

| Construct | Measure | Timing | Role |
|---|---|---|---|
| **Task success** | Correct product ID within 120 s | Per trial | Primary |
| **Completion time** | Log ms to correct product page; censored at 120 s | Per trial | Primary |
| **Navigation errors** | Wrong department/subcategory/leaf; backtrack; category switch | Per trial | Primary (composite) / secondary (components) |
| **Cognitive load** | Raw NASA-TLX (6 items) | Per block | Primary |
| **Perceived usability** | UMUX-Lite (2 items) per block; SUS (10 items) per menu at end | Block / end | Primary |
| **Disorientation / control** | 4-item scale (“I felt lost,” “I knew where I was,” “I could predict where to go,” “I felt in control”) | Per block | Primary |
| **Engagement / boredom** | 4-item short engagement scale; boredom item | Per block | Secondary |
| **Preference** | Ranking of six menus; forced choice for each state | End | Secondary |
| **Process** | Clicks, path efficiency, menu open/close, hover dwell, first-click correctness, search attempts | Per trial | Secondary / mediation |

## 8.9 Analysis plan

The primary analysis is a set of mixed-effects models with menu, state, and their interaction as fixed effects, plus trial position, difficulty, and target set as covariates. Participant and target are crossed random intercepts. A random

slope for menu by participant is included if the model converges and the random-effect structure is supported by a likelihood-ratio test; otherwise the simpler model is reported with a sensitivity analysis.

$$\text{Success: } \operatorname{logit} P(y_{ijk} = 1) = \beta_0 + \beta_{\text{menu(j)}} + \beta_{\text{state(k)}} + \beta_{\text{menu}\times\text{state(jk)}} + \beta_{\text{trial}} + \beta_{\text{difficulty}} + u_i + v_{\text{target}}$$

$$\text{Time: } \log(T_{ijk}) = \alpha_0 + \alpha_{\text{menu(j)}} + \alpha_{\text{state(k)}} + \alpha_{\text{menu}\times\text{state(jk)}} + \alpha_{\text{trial}} + \alpha_{\text{difficulty}} + a_i + b_{\text{target}} + \varepsilon_{ijk}$$

- **Success** is analysed with a binomial GLMM ( `glmer` , logit link). Report odds ratios with 95% CIs. The interaction is tested with a likelihood-ratio test against a model without the interaction; a parametric bootstrap provides a second p-value if the asymptotic test is questionable.
- **Time** is analysed with a linear mixed model on the log scale. Failures are right-censored at 120 s; a Cox frailty model (participant random effect) is the pre-registered sensitivity analysis. Report geometric means and percentage differences.
- **Errors** are counts; use a negative-binomial GLMM with an offset for opportunities. Report incidence-rate ratios.
- **Load, usability, disorientation** are analysed with LMMs (or ordinal models for individual items). Report estimated marginal means with 95% CIs.
- **Multiplicity:** the omnibus interaction is the gate. If it is not significant, the four focused contrasts are reported as exploratory with FDR-adjusted p-values. If it is significant, the four contrasts are tested with Holm correction and reported alongside the full matrix.
- **Effect sizes:** odds ratios, percentage-point differences in success, Cohen's $d_z$ for within-subject time contrasts, and incidence-rate ratios. Report precision, not just p-values.
- **Model checks:** residual plots, overdispersion, singularity, influence diagnostics; report if a simpler random-effects structure was used.
- **Sensitivity:** exclude C (click-depth confound); analyse A/B/D/E/F only; per-protocol state labels; exclude participants who guessed the hypothesis; exclude the first trial of each block (learning); include input modality as a covariate.

## 8.10 From matrix to policy

The output of Study 1 is not a single "best menu." It is a 6 × 4 matrix with uncertainty. The policy is selected with a pre-registered, transparent rule:

1. **Guardrail:** for each state, retain menus whose success rate is within 5 percentage points of the best menu in that state (non-inferiority margin). If no menu is clearly best, use the global best menu for that state.
2. **Efficiency:** among retained menus, choose the one with the shortest geometric mean completion time.
3. **Errors:** if two menus are within 10% on time, choose the one with fewer navigation errors.
4. **Experience:** if still tied, choose the higher UMUX-Lite / lower NASA-TLX.
5. **Uncertainty:** require a posterior probability ≥ 0.90 that the chosen menu is best on the primary outcome, or a clear non-inferiority result. Otherwise default to the global best menu and flag the state as "no reliable adaptation."
6. **Simplicity:** if the chosen menu is within the guardrail of the global best, prefer the global best to avoid switching costs.

A Bayesian hierarchical model ( `brms` ) with regularising priors on the 24 cell effects provides posterior probabilities and shrinks extreme cell estimates. The frequentist models remain the primary inferential analysis; the Bayesian model is used for policy selection and for communicating uncertainty. A utility-based sensitivity analysis (weights on success, time, errors, usability) checks whether the policy changes under reasonable alternative weights.

## 8.11 Expected patterns and interpretation

| Pattern | Interpretation |
|---|---|
| **C wins in most states** | Breadth and click savings dominate; the practical recommendation is a mega menu, and emotion-contingent structure adds little. This is a strong applied finding even if it weakens the adaptation story. |
| **Different winners in different states** | The core hypothesis is supported; proceed to Study 2 with the state-specific policy. The size of the differences determines whether adaptation is worth the switching cost. |
| **E/F win only under boredom** | Adaptation has a narrow, engagement-focused niche. The paper should frame it as a targeted intervention, not a universal policy. |
| **B wins under confusion** | Persistence and explicit labels matter more than breadth. This supports wayfinding theory and has direct design implications. |
| **No differences** | Either the task is too easy (ceiling), the induction is weak, or menu structure is robust to mild affect. Check manipulation and difficulty before interpreting as a null. |
| **Speed–accuracy trade-off** | E/F may be faster but less accurate; C may be accurate but visually overwhelming. Report the trade-off rather than a single winner; the policy guardrail handles it. |

## 8.12 Threats specific to Study 1

- **Induction failure or fade:** manipulation checks, re-induction, CACE analysis, and a dose-response model using continuous ratings.
- **Demand characteristics:** broad cover story, no mention of "emotion and menus," suspicion check, sensitivity analysis excluding guessers.
- **Order and learning:** Latin squares, practice trials, trial-position covariate, first-trial exclusion sensitivity.
- **Fatigue:** 90-min cap, breaks, pilot timing, fatigue rating; if fatigue is high, split into two sessions in a follow-up.
- **Menu confounds:** parity contract, click-depth mediation, matched-depth sensitivity, colour counterbalancing, animation control.
- **Task/target confounds:** unique targets, balanced difficulty, target rotation, crossed random effects.
- **Ceiling effects:** difficulty calibration in the pilot; if success > 95% overall, increase difficulty or use time/errors as the primary outcome.

## 8.13 Deliverables

A 6 × 4 matrix of success, time, errors, load, usability, and disorientation with 95% credible intervals; posterior probabilities for each menu being best in each state; a frozen policy table with a "no reliable adaptation" flag where appropriate; a fully reproducible dataset and analysis script; and a pre-registration update before Study 2.

# 9 Study 2 — causal test of adaptation

Study 1 can show that menus differ across states. It cannot show that **using** that mapping helps. Study 2 freezes the mapping and tests the causal claim against three counterfactuals: the best static menu, a deliberately mismatched menu, and a sham switch that changes nothing but announces a change.

## 9.1 Objective and logic

The core comparison is **matched vs static**. If the matched condition is better, the mapping has applied value. But a gain could come from mere change, novelty, or attention to the interface rather than from matching. The **mismatched** condition controls for change without fit; the **sham** condition controls for the announcement of change without any actual change. The four conditions form a 2 × 2 logic:

| Condition | Menu changes? | Change is matched to state? | What a gain would mean |
|---|---|---|---|
| **Static** | No | — | Baseline: the best single menu. |
| **Matched** | Yes | Yes | Fit + change. The target effect. |
| **Mismatched** | Yes | No | Change without fit. If it also helps, the effect is novelty/attention. |
| **Sham** | No (banner only) | — | Announcement without change. If it helps, the effect is demand/attention. |

## 9.2 Design

| Factor | Levels | Type | Notes |
|---|---|---|---|
| **Adaptation condition** | Static, matched, mismatched, sham | Within-subject | Four conditions per state; order balanced by a 4 × 4 Latin square. |
| **Emotional context** | Happy, frustrated, confused, bored | Within-subject | Four blocks; block order balanced across participants. |
| **Trial** | 1 per condition per state | Within-subject | 16 main trials; 4 practice trials; ~70 min. |
| **Target set** | 4 sets × 16 tasks | Between-participant rotation | Balances products across conditions and states. |

**Why four conditions rather than two?** A two-condition test (matched vs static) would be cheaper, but it would leave the most important alternative explanation — "any change helps" — untested. The mismatched and sham conditions are the difference between a convincing causal paper and a demo. If resources are tight, the sham condition can be dropped first; the mismatched condition is essential.

## 9.3 Conditions in detail

| Condition | Menu shown | Transition message | Participant control |
|---|---|---|---|
| **Static** | The global best menu from Study 1, held constant across all four states. | "The interface will remain the same for this task." | No switch; a "dislike" button is available and logged. |

| | | | |
|---|---|---|---|
| | is used and the trial is marked concordant. | "The interface will adjust to support the current task." | No revert in Study 2 (to isolate the effect); a "dislike" button is logged. |
| **Mismatched** | A menu assigned by a pre-registered derangement of the state→menu mapping: happy→bored's menu, frustrated→happy's menu, confused→frustrated's menu, bored→confused's menu. If the deranged menu equals static or matched, shift to the next menu in a fixed order. | "The interface will adjust to support the current task." (Same wording as matched.) | Same as matched. |
| **Sham** | The static menu, unchanged. A brief banner and a 400-ms highlight animation suggest an adjustment, but the layout is identical. | "The interface will adjust to support the current task." | Same as matched. |

**Handling concordance**

If the frozen policy selects the same menu as the static baseline for a state, matched and static are identical in that state. That state is **concordant** and is excluded from the matched-vs-static contrast; it is still analysed descriptively. The paper reports the number of discordant states. If fewer than two states are discordant, H2 cannot be tested and the paper reports the mapping plus the mismatched/sham results as the causal evidence. Do not secretly change the static baseline after seeing the data.

## 9.4 Participants and sample size

**Target N = 80** (full) or **N = 60** (reduced). The full sample provides ~90% power for a standardized matched-vs-static difference of d = 0.40 on the condition mean, and ~75% for d = 0.30, assuming a within-participant correlation of .50 across the four state observations. The reduced sample is powered for d ≈ 0.40. The pilot/Study 1 variance estimates replace these assumptions before the final N is fixed. Inclusion and exclusion criteria are the same as Study 1, with no overlap of participants.

## 9.5 Procedure

| Min | Step | Detail |
|---|---|---|
| **0–10** | Consent, demographics, baseline | Same screening as Study 1; baseline affect; explain that the interface may change between tasks. |
| **10–20** | Instructions and practice | Four practice trials, one per condition, using neutral practice targets. Explain the transition banner and the "dislike" button. Do not disclose the matching hypothesis. |
| **20–32** | Block 1: induction + 4 trials | 2.5-min induction, 1-min check, then four trials (static, matched, mismatched, sham) in a balanced order. Each trial begins with a 5-s transition screen. Post-block check + NASA-TLX + UMUX-Lite. |
| **32–44** | Block 2 | Next state in the Latin-square order; neutral reset before the block. |
| **44–56** | Block 3 | Same. |
| **56–68** | Block 4 | Same; final post-block check. |
| **68–78** | Questionnaires and debrief | Perceived change, understanding, control, trust, disruption, preference; suspicion check; debrief. |

**Trial sequence:** task instruction (5 s) → transition screen (5 s; condition-specific wording) → menu appears → participant finds the product (120-s cap) → neutral feedback → 3-s inter-trial interval. The transition screen is the only place where the condition is announced. The menu itself is visible immediately after.

## 9.6 Outcomes and contrasts

| Outcome | Measure | Contrast / decision rule |
|---|---|---|
| **Task success** | Correct product within 120 s | Matched vs static: non-inferiority margin 5 pp; matched vs mismatched: superiority; matched vs sham: superiority. |
| **Completion time** | Log ms to correct product; censored at 120 s | Matched vs static: superiority (faster); matched vs mismatched: superiority; report geometric mean ratio. |
| **Navigation errors** | Wrong branch/leaf, backtrack, category switch | Matched vs static and vs mismatched: fewer errors; negative-binomial model. |
| **Cognitive load** | NASA-TLX after each block | Matched vs static: lower load; matched vs mismatched: lower load. |
| **Disorientation / control** | 4-item scale after each block | Matched vs static: less disorientation; matched vs mismatched: less disorientation. |
| **Switching cost** | First trial after a change vs stable trial | Switch cost > 0 on time/errors; test whether it is offset by the matched gain. |
| **Perceived disruption** | 3 items: noticed, understood, disruptive | Matched vs sham: does the change feel disruptive even when it helps? |
| **Trust / acceptance** | 4 items; “dislike” clicks | Matched vs static: trust and acceptance; mismatched vs matched: lower trust. |

## 9.7 Analysis

The primary model is a mixed-effects model with condition, state, and condition × state as fixed effects, plus period, sequence, and target set as covariates. Participant and target are crossed random intercepts. Planned contrasts are matched vs static, matched vs mismatched, matched vs sham, and static vs mismatched. The matched-vs-static success contrast is a non-inferiority test with a 5-percentage-point margin; time and errors are superiority tests. Report the full condition × state matrix even if the interaction is not significant.

- **Carry-over:** include period and sequence; test condition × period; if carry-over is present, analyse only the first period or use a washout model.
- **Concordance:** exclude concordant states from the matched-vs-static contrast; report the number of discordant states and the per-state results.
- **Per-protocol:** if a participant clicks “dislike” or does not attend to the transition, the trial remains in the ITT analysis; a per-protocol analysis excludes trials with no detectable menu inspection.
- **Mediation:** test whether the matched effect is mediated by switching cost, perceived disruption, cognitive load, and disorientation. Use bootstrapped indirect effects with 5,000 resamples.
- **Equivalence:** if matched and static are statistically indistinguishable, report an equivalence test against a pre-specified margin (5 pp success; 10% time) rather than claiming “no difference.”
- **Bayesian:** a hierarchical model with regularising priors provides the posterior probability that matched > static and that matched > mismatched; use it for interpretation, not for p-hacking.

## 9.8 Interpretation scenarios

| Pattern | Interpretation |
|---|---|
| **Matched > static, matched > mismatched, matched > sham** | Strong evidence for emotion-contingent structural adaptation. The effect is not mere change or announcement. |
| **Matched > static, but mismatched ≈ matched** | Change/novelty/attention drives the gain; matching is not necessary. The practical recommendation is to vary the interface, not to personalise it. |
| **Matched ≈ static, matched > mismatched** | Matching helps relative to a bad fit, but the static best menu is already good enough. Adaptation has little practical value. |
| **Matched < static** | Switching costs and disruption dominate. The policy should be "keep the stable best menu" unless the state is extreme and the gain is large. |
| **Sham ≈ matched** | Demand/announcement effects. The study cannot claim a structural benefit; redesign the control and report the null. |

## 9.9 Threats specific to Study 2

- **Demand characteristics:** the transition banner makes the manipulation obvious. Use a broad cover story, do not disclose the matching hypothesis, and include a suspicion check; the sham condition helps quantify announcement effects.
- **Condition visibility:** participants can see which menu appears. That is unavoidable; the question is whether they can infer the hypothesis. The post-experiment interview probes this.
- **Concordance:** matched and static may coincide; pre-register the handling and report the number of discordant states.
- **Carry-over:** a menu change in one trial may affect the next. Use a neutral reset between conditions and include period/sequence; the first-trial-after-switch analysis quantifies the cost.
- **Learning:** four conditions × four states means participants see 16 menus changes. Practice and counterbalancing reduce learning; the trial-position covariate absorbs the rest.
- **No revert:** Study 2 removes revert to isolate the causal effect. This is a deliberate limitation; Study 3 restores user control. The ethics protocol must explain that the change is temporary and the participant can stop at any time.

## 9.10 Deliverables

A causal estimate of matched vs static, matched vs mismatched, and matched vs sham; a switching-cost estimate; a mediation model; a condition × state matrix; and a policy recommendation that states when adaptation is worth the disruption and when it is not.

# 10 Study 3 — end-to-end system with an imperfect estimator

Study 2 uses known, induced states. A real system must infer the state from noisy signals. Study 3 tests whether the adaptation still helps when the state estimate is uncertain, and it identifies the accuracy at which adaptation stops paying for itself.

## 10.1 Objective and design logic

The study compares a **static** condition (the global best menu) with a **hybrid adaptive** condition that uses a frozen estimator, a confidence threshold, and a safe transition policy. The estimator is trained on Study 1 and validated with participant-grouped cross-validation; it is not trained on Study 3 outcomes. The design is a within-subject crossover: four emotion blocks, each with one static trial and one hybrid trial in counterbalanced order, plus a short free-browsing task. This yields paired comparisons within each state while keeping the session under 75 minutes.

| Factor | Levels | Type | Notes |
|---|---|---|---|
| **System condition** | Static, hybrid adaptive | Within-subject | ABBA/BAAB order within each state; 8 main trials. |
| **Emotional context** | Happy, frustrated, confused, bored | Within-subject | Four blocks; block order Latin-square balanced. |
| **Free browsing** | 5 min per condition | Within-subject | Exploratory task with no single correct target; measures engagement and product views. |
| **Estimator** | Frozen model from Study 1 | Between-session | No outcome leakage; calibration on held-out participants. |

## 10.2 The estimator

The estimator predicts the participant's **self-reported dominant state** at the start of a task, using (a) the block-level self-report check-in and (b) interaction features from the previous task. This is a realistic "hybrid" system: it asks once, then updates from behaviour. It does not use webcam, facial expression, or physiological sensors.

| Feature family | Examples | Rationale |
|---|---|---|
| **Temporal** | Time since last click, pause duration, trial elapsed time, time to first click, idle ratio. | Confusion and boredom produce different pause patterns; frustration produces rapid clicking. |
| **Click / motor** | Click count, click rate, repeated clicks on the same element, dead clicks, mouse speed, acceleration, curvature, distance travelled, entropy. | Frustration is associated with fast, repetitive, less precise pointing; boredom with long idle periods. |
| **Navigation** | Backtracks, category switches, menu open/close count, depth reached, path length, first-click correctness, search attempts. | Confusion is the signature of inefficient wayfinding; happiness may show more exploration. |
| **Task / context** | Task difficulty, target department, menu condition, trial number, previous state, block order. | Controls for task-driven variation; prevents the model from learning the task instead of the state. |
| **Self-report** | SAM valence/arousal, four discrete items at block start and random micro-checks. | Anchors the label; provides a prior for the first trial; not available continuously. |

**Model:** a penalised multinomial logistic regression (L2, class-balanced) is the primary model because it is interpretable and calibratable. Gradient boosting (XGBoost/LightGBM) is a secondary model to check for non-linear gains. The label is the four states plus a `mixed` class; the model outputs a probability distribution. Calibration is checked with reliability diagrams and the Brier score; temperature scaling is fitted on held-out folds only.

**Validation:** leave-one-participant-out cross-validation on Study 1. Report macro-F1, balanced accuracy, per-class recall, the confusion matrix, and calibration. Compare with a majority-class baseline and with an "oracle" model that knows the induced state. The gap between the estimator and the oracle is the inference cost.

## 10.3 Transition policy

A prediction is not an instruction. The system changes the menu only when five conditions hold:

1. **Confidence:** the calibrated probability of the predicted state is ≥ 0.70 (tuned on Study 1, frozen before Study 3).
2. **Expected gain:** the predicted utility gain of the target menu over the current menu exceeds a pre-registered minimal meaningful gain (e.g., 5% of the current expected utility).
3. **Dwell:** the current menu has been used for at least two trials, so the system does not oscillate.
4. **Boundary:** the switch happens between tasks, not mid-route.
5. **User control:** the banner offers "Keep current" and "Switch back"; a revert is logged and respected for the rest of the block.

If confidence is below threshold or the model sees an out-of-distribution feature pattern, the system keeps the current menu and logs "no adaptation." This fallback is a feature, not a failure: a system that abstains when uncertain is safer than one that always switches.

## 10.4 Participants and sample size

**N = 72** (full) or **N = 24–48** (feasibility). The full sample gives ~80% power for a paired difference of d = 0.35 on the hybrid-vs-static condition mean and a 95% CI half-width of about ±0.06 for a macro-F1 near .55 (assuming four check-ins per participant). The reduced sample is a feasibility study, not a confirmatory estimator evaluation; report precision and do not overclaim. Inclusion/exclusion are the same as Study 1, with no overlap.

## 10.5 Procedure

| Min | Step | Detail |
|---|---|---|
| **0–10** | Consent, baseline, instructions | Explain that the interface may adapt; the participant can keep the current layout or switch back. Demonstrate the banner on a practice task. |
| **10–18** | Practice | Two practice trials (static and hybrid) with neutral targets; verify that the estimator's check-in and banner are understood. |
| **18–30** | Block 1: happy | Induction + check; static trial and hybrid trial in ABBA order; 2-min free browse; post-block questionnaires. |
| **30–42** | Block 2: frustrated | Same structure; neutral reset between blocks. |
| **42–54** | Block 3: confused | Same. |
| **54–66** | Block 4: bored | Same. |
| **66–78** | Questionnaires and debrief | Trust, perceived intelligence, control, disruption, preference, open-ended interview; debrief including how the estimator works. |

**Free browsing:** "Browse the store for five minutes and look at any watches you would consider buying. There is no correct answer." Measures: products viewed, categories explored, time to first product view, depth of exploration, self-reported engagement, and satisfaction. The static and hybrid free-browse sessions are counterbalanced. This tests whether adaptation helps exploration, not just known-item search.

## 10.6 Outcomes

| Level | Measures | Purpose |
|---|---|---|
| **Estimator** | Macro-F1, balanced accuracy, per-class recall, confusion matrix, Brier score, calibration slope, expected calibration error. | Does the system know the state? Where does it fail? |
| **Navigation** | Success, time, errors, backtracks, path efficiency, switch cost. | Does adaptation improve the task? |
| **Experience** | NASA-TLX, UMUX-Lite, disorientation, trust, perceived intelligence, control, satisfaction. | Does adaptation feel helpful or disruptive? |
| **Adaptation behaviour** | Switch offered, accepted, reverted, ignored; confidence at switch; reason; time since last switch. | Is the policy stable and acceptable? |
| **Free browsing** | Products viewed, categories explored, time to first product, engagement, satisfaction. | Does adaptation help exploration? |
| **Break-even** | Simulated utility under confusion matrices of varying accuracy. | How accurate must the estimator be? |

## 10.7 Analysis

- **Estimator:** participant-grouped cross-validation; report the mean and 95% CI across folds; compare with the majority baseline and the oracle. A model that cannot beat the baseline by a meaningful margin is reported as a failure, not hidden.
- **System effect:** paired mixed-effects models with condition, state, and condition × state; participant and target random intercepts; period and sequence covariates. Primary contrast: hybrid vs static on success (non-inferiority) and time/errors (superiority).
- **Confidence stratification:** split hybrid trials into high-confidence and low-confidence predictions; test whether the benefit is concentrated in high-confidence trials. This is the key mechanism test.
- **Switch cost:** compare the first trial after an accepted switch with stable trials; estimate the cost and the number of trials needed to recover.
- **Break-even simulation:** use the Study 1 6 × 4 matrix as the ground truth. Generate synthetic predictions with confusion matrices at 40%, 55%, 70%, 85%, and 100% accuracy; apply the same confidence/hysteresis policy; compute expected success, time, and utility. Plot utility against accuracy and identify the break-even point where hybrid = static. This can be done without new participants and is a standalone contribution.
- **Decision curve:** net benefit of adaptation across confidence thresholds; choose the threshold that maximises net benefit on calibration data, then freeze it for Study 3.
- **Qualitative:** interview data on trust, notice, and perceived intelligence; thematic analysis; use quotes to illustrate, not to replace the quantitative test.

## 10.8 Interpretation scenarios

| Pattern | Interpretation |
|---|---|
| **High estimator accuracy; hybrid > static; close to oracle** | The system works end-to-end. The paper can recommend deployment with a confidence threshold and user control. |
| **High accuracy but hybrid ≈ static** | Switching costs or task-boundary friction erase the gain. The bottleneck is the transition policy, not the estimator. |
| **Low accuracy; hybrid ≈ static or worse** | The estimator is the bottleneck. Report the break-even accuracy and the features that would be needed to reach it. |
| **High revert rate** | Users prefer stability or do not trust the system. The design recommendation is opt-in adaptation, not automatic switching. |
| **Benefit only in free browsing** | Adaptation supports exploration but not known-item search. This is a useful boundary condition. |

## 10.9 Threats specific to Study 3

- **Overfitting / leakage:** participant-grouped CV, frozen model, no use of Study 3 outcomes for training; report the exact feature pipeline and seeds.
- **Small N for a 5-class classifier:** report CIs and per-class support; collapse rare classes if necessary; do not report a single accuracy without the confusion matrix.
- **Self-report reactivity:** check-ins at block boundaries and random micro-checks, not after every click; report whether the check-in itself changed behaviour.
- **Switching cost / novelty:** ABBA order, stable-trial comparison, sham-like static banner, and a first-trial-after-switch analysis.
- **Demand:** participants may try to "help" the adaptation by acting frustrated. The cover story and the behavioural feature set make this difficult, but the suspicion check and a sensitivity analysis excluding guessers are needed.
- **Deployment:** lab-only, short session, fictional store; external validity is limited. A field study is future work.

## 10.10 Deliverables

A validated estimator with calibration and confusion matrix; a hybrid-vs-static effect estimate; a switch-cost estimate; a break-even accuracy curve; a transition-policy specification; and an open, reproducible pipeline that maps event logs to state probabilities and adaptation decisions.

# 11 Measures and instruments

The measurement plan balances three demands: capture the constructs precisely, keep repeated measurement short enough not to destroy the task, and avoid confounding the emotion manipulation with the outcome measures. The last point is easy to miss: the standard NASA-TLX includes a frustration subscale, which would overlap with the frustration induction. The plan therefore uses a workload score that excludes that subscale.

## 11.1 Core instruments

| Construct | Instrument / items | Timing | Scale | Role / reliability target |
|---|---|---|---|---|
| **Valence / arousal** | Self-Assessment Manikin (SAM), 2 pictorial items (Bradley & Lang, 1994) | Baseline, post-induction, post-block, 2 micro-checks per block | 9-point | Manipulation check and continuous covariate; no internal consistency expected (2 different dimensions). |
| **Discrete emotion** | 4 items: happy, frustrated, confused, bored; 7-point "not at all" to "extremely"; plus "none of these" | Baseline, post-induction, post-block | 7-point | Primary state label; pilot checks comprehension and discriminant validity. |
| **Cognitive load** | NASA-TLX workload: mental demand, temporal demand, effort, performance (reverse-scored); **frustration subscale excluded** (Hart & Staveland, 1988) | After each block | 0–100 visual analogue or 21-point | Primary mediator/outcome; $\alpha \geq .70$ in pilot. |
| **Perceived usability** | UMUX-Lite (2 items) after each block; SUS (10 items; Brooke, 1996) per menu at the end | Block / end | 7-point / 5-point | Primary usability; SUS $\alpha \geq .70$; UMUX-Lite correlation with SUS $\geq .70$ in pilot. |
| **Disorientation** | 4 items: "I felt lost," "I knew where I was," "I could predict where to go," "I felt in control" (2 reverse-scored) | After each block | 7-point | Primary wayfinding measure; $\alpha \geq .75$. |
| **Engagement** | 4-item short form adapted from the User Engagement Scale (O'Brien et al., 2018): focused attention, perceived usability, aesthetic appeal, reward | After each block | 7-point | Secondary; $\alpha \geq .70$. |
| **Trust / perceived intelligence** | 4 items: "The interface understood my needs," "The change was appropriate," "I trusted the system," "The interface felt intelligent" | Study 2/3, after each condition | 7-point | Secondary adaptation measure; $\alpha \geq .75$. |
| **Adaptation experience** | Notice, understanding, disruption, acceptance, revert intention; behavioural revert/keep-current counts | Study 2/3, after each block | 7-point + counts | Secondary; single items reported separately and as a composite if $\alpha$ supports it. |
| **Preference** | Rank the six menus; forced choice for each state ("Which would you want if you felt this way?") | End of Study 1/2 | Rank / choice | Secondary; descriptive, not a substitute for performance. |

| | | | | |
|---|---|---|---|---|
| | for cognition; 4-item ambiguity tolerance | Baseline | Mixed | Exploratory moderators; keep short to avoid burden. |
| **Suspicion / strategy** | Open-ended: "What did you think the study was about?" "Did you notice any pattern in the interface?" | Debrief | Open | Demand-characteristic check; sensitivity analysis excluding correct guessers. |

**Why the NASA-TLX frustration subscale is excluded**

The standard NASA-TLX has six subscales, one of which is "frustration level." Using it after a frustration induction would make the load measure partly a manipulation check, inflating the association between state and load. The plan uses a **four-item workload score** (mental demand, temporal demand, effort, performance) and treats frustration as an emotion variable, not a workload variable. If a full NASA-TLX is reported for comparability, the frustration subscale is reported separately and never included in the workload composite.

## 11.2 Self-report timing and reactivity

Repeated self-report can interrupt navigation and make participants aware of their state. The design uses three levels of measurement:

- **Block-level:** full SAM + discrete items before and after each emotion block; NASA-TLX, UMUX-Lite, disorientation, and engagement after each block. This is the primary self-report.
- **Micro-check:** a 2-item valence/arousal check after trials 3 and 6 of each block (not after every trial). This provides trial-level anchors without constant interruption.
- **End-of-session:** SUS per menu, preference ranking, trust/adaptation items, suspicion check, and interview. These are not repeated per trial.

For the estimator, the block-level label is the target; micro-checks are used only to assess fade. The model is trained on Study 1 block labels and features from the preceding trials. This avoids using a trial-level self-report that would be too sparse to train on.

## 11.3 Reliability and validity plan

| Check | Method | Criterion / action |
|---|---|---|
| **Internal consistency** | Cronbach's α and McDonald's ω for multi-item scales in the pilot. | α/ω ≥ .70; if lower, inspect item-total correlations and remove or revise the item before the main study. |
| **Comprehension** | Cognitive interviews with 5 participants; think-aloud on each item. | No item misunderstood by more than one participant; revise wording. |
| **Discriminant validity** | Correlations among the four discrete emotion items; expected moderate, not r > .85. | If two states are indistinguishable (r > .85), merge or revise the induction; report the overlap. |
| **Manipulation validity** | Dominant-state rule; change from baseline; dose-response with intensity. | ≥ 60% intended dominant; intensity ≥ 5/7; otherwise revise. |
| **Convergent validity** | UMUX-Lite vs SUS; NASA-TLX vs disorientation; expected moderate correlations. | r ≥ .50 for convergent pairs; if not, report and interpret cautiously. |
| **Reactivity** | Compare trials with and without a preceding micro-check. | No meaningful difference in time/success; if present, report and adjust. |

## 11.4 Optional eye-tracking extension

Eye tracking is not required for the core claim, but it can explain *why* a menu helps. The folder's Chiu and Yang (2024) paper provides a useful AOI/metric template. The extension uses a screen-based tracker (e.g., Tobii Pro

Spectrum or Fusion, 120 Hz), a 9-point calibration, and a fixation filter pre-registered as I-VT with a 30°/s velocity threshold and 100-ms minimum fixation.

| AOI / metric | Definition | Hypothesis |
|---|---|---|
| **Target department AOI** | Bounding box of the correct department in each menu. | Faster time to first fixation in menus that make the category salient (C, E, B). |
| **Target leaf AOI** | Bounding box of the correct leaf at the final level. | More direct scanpath and fewer fixations in C (all leaves visible) and E (icon salience). |
| **Menu region AOI** | Top bar, dropdown, flyout, panel, sidebar, tile grid, hub, rings. | Dwell distribution reveals whether users search the hierarchy or the product area. |
| **Fixation count / duration** | Number and total duration of fixations per AOI. | Lower load in the matched menu; higher in dense panels under frustration. |
| **Scanpath length** | Sum of saccade amplitudes between fixations. | Shorter, more direct paths in familiar/spatially stable menus. |
| **AOI transitions** | Number of transitions between menu levels and product area. | More transitions indicate backtracking/uncertainty (confusion). |
| **Pupil diameter (optional)** | Mean pupil size during the trial, controlling for luminance. | Arousal/load; interpret cautiously because luminance differs across menus. |

Eye-tracking analysis uses LMMs with menu, state, and their interaction; participant random intercepts; AOI size and luminance as covariates. Trials with calibration error > 1.5° or > 20% missing gaze are excluded. The extension requires separate ethics approval because gaze data are more identifying than click logs; store raw gaze separately and do not share it publicly.

## 11.5 Data-quality checks

- **Attention:** one embedded instruction check (“Press 3 if you are reading this”) in the questionnaires; failure is flagged, not automatically excluded.
- **Straight-lining:** long runs of identical responses; flag and inspect.
- **Completion time:** questionnaire completion < 2 min or > 3 SD above the mean is flagged.
- **Technical:** missing `trial_end`, duplicate events, clock drift, browser errors; automated validation on ingestion.
- **Randomisation:** server-side audit of condition sequences; no participant sees a condition twice per state.
- **Manipulation:** dominant-state rate by induction, order, and participant; report failures.
- **Bots / duplicate IDs:** for online recruitment, check IP duplicates (hashed), completion patterns, and open-ended responses; exclude only with documented evidence.

# 12 Analysis plan

The analysis is pre-registered, model-based, and effect-size focused. The goal is not to find a significant p-value for the 6 × 4 interaction; it is to estimate the matrix with enough precision to make a defensible policy decision. Where the design is underpowered for a small effect, the paper reports the uncertainty rather than pretending the null is definitive.

## 12.1 Principles

1. **Pre-register** the primary outcomes, the 6 × 4 interaction, the four focused contrasts, the Study 2 contrasts, and the policy rule.
2. **Analyse as randomised** (intention-to-treat) for the primary test; report per-protocol and CACE as sensitivity analyses.
3. **Model the clustering** of trials within participants and targets; do not treat 1,728 trials as independent.
4. **Report effect sizes with 95% CIs**; exact p-values; no dichotomous "significant / not significant" language without the estimate.
5. **Correct for multiple comparisons** within families; label exploratory analyses.
6. **Freeze the mapping** after Study 1; do not revise it after seeing Study 2.
7. **Share** the anonymised data, analysis code, task bank, and materials where ethics allow.

## 12.2 Data preparation

| Step | Rule |
|---|---|
| **Event → trial table** | Collapse events by `trial_id`; compute success, time, clicks, errors, backtracks, category switches, path efficiency, first-click correctness, menu open/close count, search attempts. Validate against the raw log for 5% of trials. |
| **Time** | Use client `performance.now()`; subtract trial start; cap at 120,000 ms. Log-transform after adding 1 ms. Failures are censored, not set to 120 s in the primary LMM; the survival model handles censoring. |
| **Outliers** | Flag trials with time > 3 SD from the participant×menu×state mean after log transform; inspect, do not automatically delete. Report a sensitivity analysis excluding flagged trials. |
| **Missing data** | Missing trial outcomes are not imputed; they are analysed as failures if the trial started and no product was reached. Missing questionnaire items (< 10%) are imputed with multiple imputation (m = 20) as a sensitivity analysis; the primary analysis uses complete cases. |
| **Exclusions** | Pre-registered only: incomplete log, < 75% trials, technical failure, prior participation. Report a flow diagram with reasons. |
| **Derived variables** | Path efficiency = min clicks / actual clicks; backtrack = deeper-to-shallower transition without a leaf; category switch = change of department or subcategory after a selection. |
| **State label** | Dominant-state rule (§3.3). Mixed/neutral is retained as a category in descriptive analyses; the primary model uses the randomised state (ITT) and a sensitivity model uses the dominant label. |

## 12.3 Primary models

$$\text{Success: } \operatorname{logit} P(y_{ijk} = 1) = \beta_0 + \beta_{\text{menu(j)}} + \beta_{\text{state(k)}} + \beta_{\text{menu×state(jk)}} + \beta_{\text{trial}} + \beta_{\text{difficulty}} + u_i + v_{\text{target}}$$

$$\text{Time: } \log(T_{ijk}) = \alpha_0 + \alpha_{\text{menu(j)}} + \alpha_{\text{state(k)}} + \alpha_{\text{menu×state(jk)}} + \alpha_{\text{trial}} + \alpha_{\text{difficulty}} + a_i + b_{\text{target}} + \varepsilon_{ijk}$$

Use sum-to-zero contrasts for menu and state so that main effects are interpretable as average effects. The interaction is tested with a likelihood-ratio test (or a parametric bootstrap when the asymptotic approximation is doubtful). Estimated marginal means and pairwise contrasts are computed with `emmeans`; p-values are adjusted within the pre-registered families. For the four focused contrasts, use Holm correction; for the exploratory 6 × 4 cell comparisons, use Benjamini–Hochberg FDR and label them exploratory.

| Outcome | Model | Reporting |
|---|---|---|
| **Success** | Binomial GLMM, logit link; random intercepts for participant and target. | Odds ratios, 95% CIs, predicted probabilities by cell. |
| **Time** | LMM on log time; Cox frailty model as censoring sensitivity. | Geometric means, ratio of geometric means, 95% CIs. |
| **Errors / backtracks** | Negative-binomial GLMM with offset for opportunities. | Incidence-rate ratios, 95% CIs. |
| **NASA-TLX / UMUX-Lite / disorientation** | LMM or ordinal cumulative-link model for individual items. | Estimated marginal means, 95% CIs, standardized differences. |
| **Preference / ranking** | Descriptive; Friedman test; paired Wilcoxon with correction. | Rank distributions, Kendall's W, effect sizes. |
| **Study 2 condition effects** | Same mixed-model family with condition, state, condition × state, period, sequence. | Planned contrasts; non-inferiority and equivalence tests. |
| **Study 3 estimator** | Participant-grouped CV; multinomial logistic regression and gradient boosting. | Macro-F1, balanced accuracy, per-class recall, confusion matrix, Brier score, calibration slope. |
| **Mediation** | Structural equation model or causal mediation with bootstrapped indirect effects. | Indirect effect, 95% CI, proportion mediated. |

## 12.4 Multiplicity, equivalence, and the "no difference" problem

A non-significant interaction is not evidence that menus are equivalent. If the interaction is small and the CI is wide, the correct conclusion is "inconclusive." If the CI excludes a pre-specified meaningful effect, the correct conclusion is "equivalent within the margin." The paper therefore reports:

- The interaction estimate and its 95% CI on the log-odds / log-time scale.
- A equivalence test for the global-best menu vs the state-specific menu, with a margin of 5 percentage points on success and 10% on time.
- A "practical significance" statement: would the observed difference change the recommended menu under the policy rule?

## 12.5 Bayesian policy model

The policy model is a hierarchical Bayesian version of the primary model, fitted with `brms`. Regularising priors (normal(0, 1) on the log-odds scale; half-normal(0, 1) on random-effect SDs) shrink extreme cell estimates and produce posterior probabilities that each menu is best in each state. The policy rule in §8.10 is applied to the posterior predictive distribution. The Bayesian model is not used to "find significance"; it is used to quantify uncertainty and to avoid overfitting a 24-cell matrix with a modest sample.

## 12.6 Qualitative and process data

Retrospective think-aloud and interview data are analysed with reflexive thematic analysis. The codebook is developed inductively from a subset of transcripts, then applied to the rest; a second coder independently codes 20% of the material. Disagreements are resolved by discussion. Qualitative data are used to explain unexpected patterns, illustrate switching costs, and generate hypotheses; they are not used to replace the quantitative test. Quotes are anonymised and checked for identifying details.

## 12.7 Reporting and reproducibility

- **Flow diagram:** participants screened, enrolled, completed, excluded, analysed.
- **Descriptive table:** demographics, baseline affect, prior menu experience, colour vision, input modality.
- **Manipulation check:** dominant-state rate by induction and order; mean intensity; adverse events.
- **Primary results:** 6 × 4 matrix with estimates and 95% CIs; interaction test; focused contrasts; effect sizes.
- **Secondary results:** usability, load, disorientation, engagement, preference; mediation.
- **Study 2/3:** condition contrasts, switch cost, estimator performance, calibration, break-even curve.
- **Transparency:** pre-registration link, analysis code, data dictionary, session logs, software versions, random seeds.

## 12.8 Analysis code skeleton (R)

```r
# Primary success model
m_success ← glmer(
  success ~ menu * state + trial_pos + difficulty +
            (1 | participant_id) + (1 | target_id),
  data = trials, family = binomial(link = "logit"),
  control = glmerControl(optimizer = "bobyqa")
)
car::Anova(m_success, type = 3)

# Time model (log scale; failures censored separately)
m_time ← lmer(
  log_time ~ menu * state + trial_pos + difficulty +
             (1 | participant_id) + (1 | target_id),
  data = trials_success, REML = FALSE
)
emmeans::emmeans(m_time, pairwise ~ menu | state, adjust = "holm")

# Errors
m_errors ← glmer.nb(
  errors ~ menu * state + trial_pos + difficulty +
           offset(log(opportunities)) +
           (1 | participant_id) + (1 | target_id),
  data = trials
)

# Censoring sensitivity
m_surv ← coxme::coxme(
  Surv(time_ms, success) ~ menu * state + difficulty + (1 | participant_id),
  data = trials
)

# Bayesian policy model
m_bayes ← brms::brm(
  success ~ menu * state + (1 | participant_id) + (1 | target_id),
  data = trials, family = brms::bernoulli(),
  prior = c(prior(normal(0, 1), class = "b"),
            prior(normal(0, 1.5), class = "Intercept"),
            prior(exponential(1), class = "sd")),
  chains = 4, iter = 4000, seed = 20261023
)
```

The code is a skeleton, not the final script. The final script is frozen before data lock, run in a container with a session-info file, and archived with the manuscript.

# 13 Sample size and power

The sample sizes in this plan come from a Monte Carlo simulation of the actual design, not from a rule of thumb. The simulation shows something important: a 6 × 4 within-subject interaction has 15 numerator degrees of freedom, so it is much harder to detect than a two-condition comparison. The plan is powered for a medium interaction, and it uses four focused contrasts to recover power for the specific predictions.

## 13.1 Simulation assumptions

| Assumption | Value | Justification / sensitivity |
|---|---|---|
| **Design** | 6 menus × 4 states, 24 trials per participant | One observation per cell; the participant intercept induces the within-subject correlation. |
| **Participant intercept SD** | 0.50 (log-time units) | Moderate individual differences; varied in sensitivity analysis from 0.3 to 0.8. |
| **Residual SD** | 1.00 | Scales the interaction effect; f is defined relative to this SD. |
| **Interaction effect SD (f)** | 0.08–0.15 | A 0.10 effect is roughly a 10% shift in the cell mean; 0.15 is a 15% shift. Pilot/Study 1 estimates replace these values. |
| **Sphericity** | Assumed; sensitivity with ε = 0.7 | Repeated-measures designs often violate sphericity; the mixed model with random slopes is more robust, but power falls. |
| **Alpha** | .05, two-sided | Primary interaction; focused contrasts use Holm correction. |
| **Replications** | 1,200 per cell | Monte Carlo error ≈ ±.014 at power = .80; adequate for planning. |

## 13.2 Study 1 — omnibus interaction (time outcome)

Table 13.1 — Power to detect the 6 × 4 interaction (Monte Carlo, 1,200 replications)

| Interaction SD (f) | N = 24 | N = 36 | N = 48 | N = 60 | N = 72 |
|---|---|---|---|---|---|
| **0.06 (very small)** | .09 | .12 | .19 | .24 | .28 |
| **0.08 (small)** | .17 | .22 | .32 | .42 | .49 |
| **0.10 (small–medium)** | .25 | .38 | .51 | .63 | .75 |
| **0.12 (medium)** | .32 | .58 | .74 | .83 | .90 |
| **0.15 (medium–large)** | .54 | .81 | .92 | .97 | .99 |

**Reading the table.** N = 48 is powered for f ≈ 0.15 (92%) but not for f = 0.10 (51%). N = 72 is powered for f ≈ 0.12 (90%) and gives 75% for f = 0.10. If the true interaction is small, even N = 72 will often produce an inconclusive result; the paper must report the CI and not claim "no effect." This is why the design also pre-registers four focused contrasts.

## 13.3 Study 1 — focused contrasts

A focused contrast compares two menus within one state (e.g., C vs A under frustration). It has 1 df and much more power than the 15-df omnibus. The table below is a paired-comparison approximation: d is the standardized difference between the two cell means, with a within-participant correlation of .50. It is illustrative; the exact power depends on the correlation and the number of trials per cell.

Table 13.2 — Approximate power for a focused two-menu contrast within one state

| Standardized difference (d) | N = 48 | N = 60 | N = 72 | N = 80 |
|---|---|---|---|---|
| **0.25 (small)** | .39 | .48 | .55 | .59 |
| **0.30 (small–medium)** | .52 | .62 | .70 | .76 |
| **0.40 (medium)** | .77 | .86 | .92 | .94 |
| **0.50 (medium–large)** | .93 | .97 | .98 | .99 |

The focused contrasts are secondary to the omnibus gate. If the omnibus is not significant, they are reported as exploratory with FDR correction. They are still useful: a strong C-vs-A effect under frustration can be a publishable finding even if the full 6 × 4 interaction is not significant.

## 13.4 Study 2 — matched vs static

Table 13.3 — Power for the matched-vs-static paired contrast (simulation, 8,000 replications, correlation .50)

| Standardized difference (d) | N = 48 | N = 60 | N = 72 | N = 80 |
|---|---|---|---|---|
| **0.25** | .39 | .48 | .55 | .59 |
| **0.30** | .52 | .62 | .70 | .76 |
| **0.40** | .77 | .86 | .92 | .94 |
| **0.50** | .93 | .97 | .98 | .99 |

N = 80 gives 94% power for d = 0.40 and 76% for d = 0.30. N = 60 gives 86% for d = 0.40 but only 62% for d = 0.30. If the expected effect is small, the reduced design is underpowered; the paper should say so and report the precision. Do not add participants after looking at the effect (optional stopping); fix N in the pre-registration.

## 13.5 Study 3 — estimator precision

Table 13.4 — Expected 95% CI half-width for a macro-F1 near .55, assuming four check-ins per participant

| N | State estimates | SE (approx.) | 95% CI half-width |
|---|---|---|---|
| **24** | 96 | .051 | ±.100 |
| **48** | 192 | .036 | ±.070 |
| **72** | 288 | .029 | ±.057 |
| **96** | 384 | .025 | ±.050 |

and cluster-bootstrap CIs, not a naive binomial interval. The table is for planning, not for inference.

## 13.6 Sample-size ladder

| Track | What you get | N (plus 20% buffer) | Timeline | Main limitation |
|---|---|---|---|---|
| **A · Minimum** | Pilot + Study 1 only; focused contrasts; descriptive matrix. | 8 + 48 = 56 (recruit ~68) | 8–10 weeks | No causal test; omnibus underpowered for small effects. |
| **B · Recommended semester** | Pilot + Study 1 (N=48) + Study 2 (N=60). | 10 + 48 + 60 = 118 (recruit ~142) | 12–16 weeks | Study 1 powered for f ≈ .15; Study 2 for d ≈ .40. |
| **C · Full paper** | Pilot + Study 1 (N=72) + Study 2 (N=80) + Study 3 (N=72). | 10 + 72 + 80 + 72 = 234 (recruit ~281) | 6–9 months | Recruitment and lab time; estimator N still modest. |
| **C+ · Eye tracking** | Full paper plus a 24-participant eye-tracking extension. | 234 + 24 = 258 (recruit ~310) | 7–10 months | Equipment, calibration, and gaze-data ethics. |

The buffer is 20% for attrition, technical failure, and pre-registered exclusions. It is not a licence to replace participants whose results are inconvenient. If recruitment is slower than expected, reduce the design (e.g., 6 × 2 states) rather than running an underpowered 6 × 4 and interpreting a null as evidence of no effect.

## 13.7 Power figure

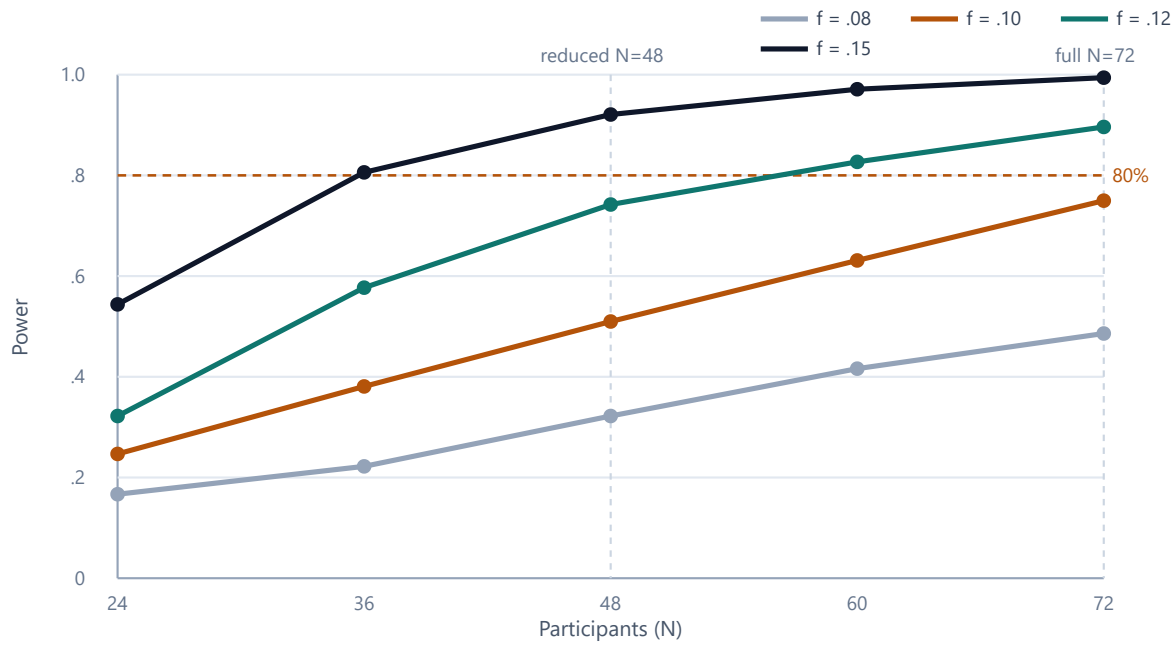



**Figure 13.1 — Power to detect the 6 × 4 interaction (time outcome).** Monte Carlo simulation, 1,200 replications per point; participant intercept SD = 0.5, residual SD = 1.0; f is the SD of the interaction effects. The vertical dashed lines mark the reduced (N = 48) and full (N = 72) Study 1 samples. A 6 × 4 interaction has 15 numerator df, so even the full sample is powered for a medium, not a small, interaction. The focused contrasts in Table 13.2 recover power for specific predictions.

## 13.8 If you cannot reach the target N

- **Reduce the state factor:** run a 6 × 2 design (happy vs frustrated) with N = 48; this halves the trials and increases power per cell. Treat the other two states as a separate exploratory study.

- **Use focused contrasts:** pre-register two or three state-specific comparisons rather than the omnibus; report the omnibus as descriptive.
- **Add trials per cell:** two targets per cell (48 trials) or a second session; this increases precision but raises fatigue and cost. Model the additional targets as a random effect.
- **Multi-site:** run the same protocol at two labs with a shared pre-registration and a site random effect.
- **Be honest:** report the achieved precision and label the study as underpowered for small effects. A well-reported inconclusive result is more valuable than an overclaimed significant one.

### The one-sentence sample-size story

For a semester project, **N = 48 in Study 1 and N = 60 in Study 2** is the practical minimum for a credible paper, powered for medium effects; for a full paper, use **N = 72 and N = 80** and add Study 3. Do not run the full 6 × 4 with N = 24 and interpret a null interaction as evidence that emotion does not matter.

# 14 Implementation plan

This section is the build contract for the prototype. It assumes a small team (one developer plus one researcher) and a six-week build before the pilot. The goal is not a production e-commerce site; it is a controlled, reproducible instrument that can run six menu architectures over one taxonomy and produce analysis-ready logs.

## 14.1 Architecture

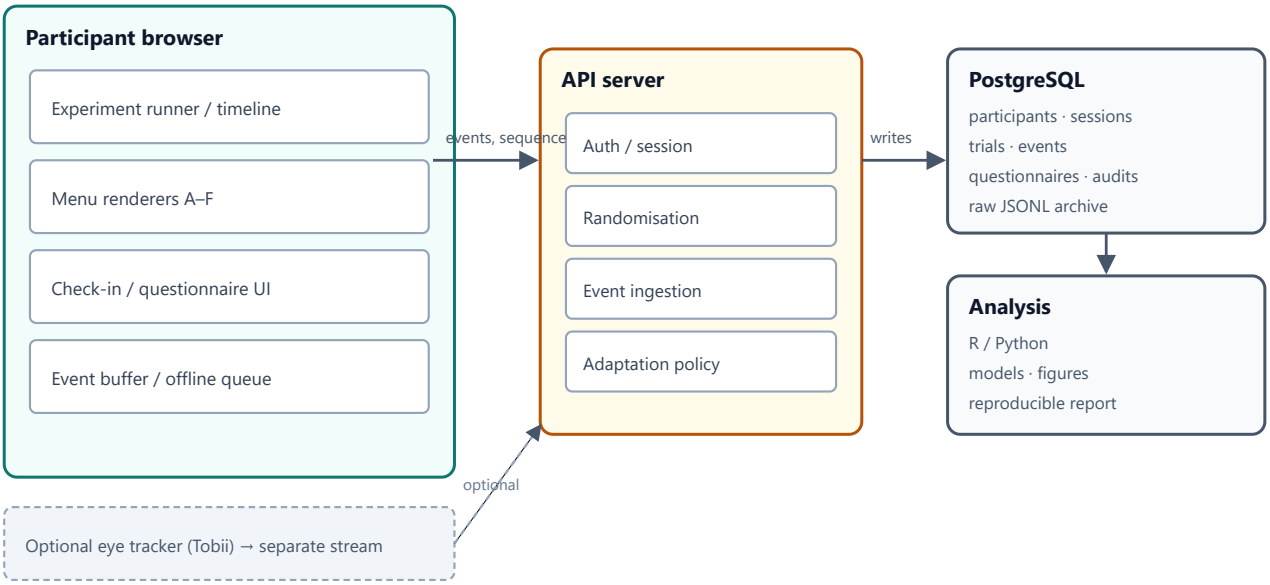



**Figure 14.1 — Recommended architecture.** The browser owns rendering and timing; the server owns randomisation and ingestion; the database owns the audit trail; the analysis pipeline reads a versioned export. The eye tracker is optional and writes to a separate stream.

## 14.2 Shared taxonomy and the renderer contract

The single most important engineering decision is to make the taxonomy a **single source of truth**. Every menu renderer reads the same JSON tree and emits the same event vocabulary. A change to a label or a product must propagate to all six menus or fail the build.

```ts
export type MenuId = "A" | "B" | "C" | "D" | "E" | "F";

export interface MenuNode {
  id: string;                 // stable, e.g. "sport.prodiver.titanium"
  label: string;
  level: 1 | 2 | 3;
  parentId: string | null;
  productId?: string;          // only for leaves
  colour?: string;             // department colour token
  icon?: string;               // icon name, same across menus
}

export interface MenuRendererProps {
  menuId: MenuId;
  nodes: MenuNode[];
  visibleLevel: 1 | 2 | 3;
  onOpen: (nodeId: string) => void;
  onSelect: (nodeId: string) => void;
  onBack: () => void;
  log: (event: LogEvent) => void;
  disabled?: boolean;
}

export interface MenuRenderer {
  mount(el: HTMLElement, props: MenuRendererProps): void;
  update(props: Partial<MenuRendererProps>): void;
  destroy(): void;
}
```

Each menu is a separate component that implements this contract. The runner mounts one renderer at a time and logs every open/select/back event with the same schema. The taxonomy package also exports a `validateTaxonomy()` function that checks: four departments; three L2 per L1; six L3 per L2; unique IDs; unique product descriptions; no orphan nodes.

## 14.3 Randomisation service

Randomisation is server-side and auditable. The service pre-generates balanced sequences from a seeded PRNG and assigns each participant to the next sequence in a stratified block. The client never chooses a condition. The audit table stores the sequence ID, block order, menu orders, target set, and a hash of the seed.

```
// Pseudocode — server-side assignment
sequences = generateBalancedDesign({
  blockOrders: latinSquare(4),      // ABCD, BCDA, CDAB, DABC
  menuOrders: latinSquare(6),       // 6 × 6
  targetSets: 6,
  seed: CONFIG.seed
});

onJoin(participant):
  stratum = stratify(participant.gender, participant.shoppingFrequency)
  seq = nextUnused(sequences, stratum)
  audit(participant.id, seq.id, hash(CONFIG.seed))
  return seq
```

Balance checks run nightly: each block order, menu position, and target set should be used with approximately equal frequency. If a stratum is exhausted, generate a new block rather than reusing sequences. Any deviation is logged and reported in the paper's supplementary materials.

## 14.4 Adaptation engine

The engine is deliberately conservative. It runs at task boundaries, uses a calibrated posterior, and abstains when uncertain. The pseudocode below is the specification for Study 3; Study 2 uses a frozen policy with known labels and no estimator.

```
onTaskBoundary(participant, currentMenu):
  features = extractFeatures(participant.recentTrials)
  prior = participant.blockCheckin          // 1-item state prior
  posterior = estimator.predict(features, prior)
  state = argmax(posterior)
  confidence = max(posterior)

  if confidence < 0.70:
      log("no_adaptation", reason="low_confidence")
      return currentMenu

  target = policy[state]
  gain = expectedUtility(target, state) − expectedUtility(currentMenu, state)

  if gain < MIN_GAIN or dwell < 2 or recentSwitch:
      log("no_adaptation", reason="guardrail")
      return currentMenu

  showBanner(target, reason="support_current_task")
  if user.keepsCurrent:
      log("adaptation_rejected")
      return currentMenu

  log("adaptation_accepted", state, confidence, gain)
  return target
```

The engine logs every decision, including abstentions. This is essential for the analysis: a system that adapts rarely but well is different from one that adapts often and badly, even if their average task performance is similar.

## 14.5 Testing strategy

| Test level | What is tested | Tool / gate |
|---|---|---|
| **Unit** | Taxonomy validation; route uniqueness; scoring functions; feature extraction; policy guardrails. | Vitest/Jest; 100% coverage of scoring and policy code. |
| **Integration** | Every menu × every target path; event log reconstruction; questionnaire submission; check-in flow. | Playwright; 432 paths (6 × 72) in nightly run; a 24-path smoke suite on every commit. |
| **Visual** | Screenshots of each menu at each level; no clipping; hit targets ≥ 44 px; focus states. | Playwright screenshots; manual review; pixel diff with tolerance. |
| **Timing** | Event order; `performance.now()` monotonicity; frame rate; transition duration. | Instrumented test run; 60 fps target; no long tasks > 50 ms during a trial. |
| **Data** | Schema validation; duplicate event IDs; missing `trial_end`; clock skew; export checksum. | Zod/JSON Schema at ingestion; nightly data-quality report. |
| **Randomisation** | Balance of block orders, menu positions, target sets; no duplicate condition per state. | Audit script; chi-square balance report. |
| **Accessibility** | Keyboard path; focus trap; ARIA roles; contrast; reduced motion. | axe-core; manual NVDA/VoiceOver smoke test; keyboard-only walkthrough. |
| **Load** | 20 concurrent sessions; ingestion latency; DB connections. | k6/Artillery; local lab server; error rate < 0.1%. |
| **Pilot dry run** | Full session with two lab members; timing; confusion; data export. | Sign-off checklist; freeze the build. |

## 14.6 Deployment, security, and data handling

- **Lab deployment:** a wired local server, Chrome/Edge kiosk mode, full-screen 1920 × 1080, no internet during a trial. The database runs on the same machine or a local network host; daily encrypted backups to a separate drive.
- **Cloud option:** a managed Postgres (EU region) and a static front end behind authentication; a data-processing agreement; no third-party analytics or advertising scripts.
- **Security:** HTTPS, content-security policy, no third-party scripts, rate limiting, parameterised queries, least-privilege DB roles, secrets in environment variables only (never in the repo).
- **Privacy:** pseudonymous participant IDs; no names, emails, or IP addresses in the research dataset; a separate encrypted key linking IDs to consent forms; retention according to ethics approval (e.g., 5 years) and deletion thereafter.
- **Audit:** every randomisation and adaptation decision is written to an append-only audit table; the raw event stream is archived as JSONL with a checksum.

## 14.7 Accessibility implementation

| Requirement | Implementation |
|---|---|
| **Keyboard** | Tab/Shift+Tab to move; Enter/Space to activate; Esc to close; arrow keys within grids/rings; focus returns to the trigger after close. |
| **ARIA** | Navigation landmarks; `aria-expanded`, `aria-haspopup`, `aria-current`; labelled nodes; live region for the transition banner. |
| **Focus** | Visible 2-px focus ring with 3:1 contrast; focus trap only inside modal-like panels; no focus loss on re-render. |
| **Contrast** | Text ≥ 4.5:1; UI components ≥ 3:1; colour never the sole carrier of meaning (labels + icons). |
| **Motion** | `prefers-reduced-motion` disables zoom/rotate transitions; instant state change; no autoplay audio. |
| **Input modality** | Mouse/trackpad is the confirmatory standard; keyboard is supported and logged; a keyboard-only session can be run separately. |

## 14.8 Reproducibility package

Use a monorepo with `apps/web`, `apps/api`, `packages/taxonomy`, `packages/analysis`, and `docs/`. Commit lockfiles, a Docker Compose file, a `.env.example` (no secrets), and a README that a new researcher can follow from a fresh clone to a running local session. Version the taxonomy and store its hash with every session. Archive the analysis script and a session-info file with the manuscript; deposit the anonymised data and materials in a repository with a DOI if ethics allow.

## 14.9 Build schedule

| Week | Deliverable | Exit criterion |
|---|---|---|
| 1 | Repo, taxonomy JSON, task bank, wireframes, event schema, CI. | Taxonomy validation passes; 72 products and 72 tasks generated; schema documented. |
| 2 | A/B/C renderers, experiment runner, event logging, local questionnaire flow. | All three menus reach every target; logs reconstruct paths. |
| 3 | D/E/F renderers, transitions, accessibility pass, visual regression. | All six menus pass the parity checklist; keyboard path works. |
| 4 | Randomisation service, backend ingestion, check-in UI, exports. | Balance audit passes; export checksum matches raw log. |
| 5 | Adaptation engine, estimator stub, policy guardrails, study-specific flows. | Study 1/2/3 dry runs complete; no missing events. |
| 6 | Pilot dry run, bug fixes, freeze, documentation. | Two lab members complete the full protocol; sign-off. |

### Build risk register

**Custom geometry takes longer than expected:** prototype D/F in week 1 with static SVG before wiring interactions. **Browser performance:** test 60 fps early; avoid heavy DOM for the mega panel. **Data loss:** offline queue plus nightly backup; test a network drop. **Randomisation bug:** server-side audit and a dry-run balance check. **Accessibility debt:** keyboard tests from week 2, not at the end. **Estimator leakage:** freeze the pipeline and use participant-grouped CV; no Study 3 outcomes in training.

# 15 Ethics, privacy, and risk

The study induces mild emotions and observes navigation behaviour. Neither is inherently harmful, but both require care: the frustration induction can feel manipulative, the confusion task can make participants feel incompetent, and the interaction logs can become identifying if handled casually. The ethics plan is therefore part of the design, not a formality at the end.

## 15.1 Approval and oversight

Obtain approval from the relevant institutional research ethics committee before recruitment or pilot data collection. The protocol, consent forms, induction scripts, questionnaires, data-management plan, and debrief must be approved. Any change to the induction, the deception, the data collected, or the sample requires an amendment. If the study is conducted across institutions, obtain approval at each site and a data-sharing agreement.

## 15.2 Informed consent

Participants receive a plain-language information sheet covering: the purpose (comparing website layouts for product finding); what they will do; the duration; the mild mood tasks; the fact that the interface may change; the data collected (clicks, timing, questionnaire responses, optionally gaze); how data are stored and shared; the right to withdraw without penalty; compensation; and contact details. Consent is recorded electronically or on paper, separate from the research data.

> **Consent must mention the mood tasks without revealing the hypothesis**
>
> "This study compares different website navigation layouts. You will complete short product-finding tasks and brief mood exercises, including a short comedy clip, a search task that may not work as expected, an ambiguous categorisation task, and a repetitive catalogue task. These are mild and temporary. You can stop at any time." This is honest about what happens without telling participants that we are testing whether frustration changes menu choice.

## 15.3 Emotion induction: risks and safeguards

| Induction | Potential risk | Safeguard | Stop / debrief |
|---|---|---|---|
| **Happiness** | Mild amusement; possible discomfort with comedy content. | Pre-screen the clip for offensive content; allow skipping; no obligation to laugh. | Stop if distress > 6/10; debrief that the clip was for mood. |
| **Frustration** | Irritation, feeling incompetent, or distrust of the researcher. | Buggy search, not personal failure; cap at three attempts; no false feedback about ability; neutral copy; experimenter support. | Immediate debrief: "The search was designed to fail; it was not your fault." Offer a break and a neutral task. |
| **Confusion** | Feeling stupid or anxious about "getting it wrong." | Instructions state there is no right answer; no feedback; the ambiguity is the point. | Reveal the ambiguity; explain the purpose; check affect before continuing. |
| **Boredom** | Frustration, fatigue, or feeling the task is pointless. | Short (90 s), clear cover story, no quota; allow a break. | Stop at 90 s; debrief; offer a stimulating break. |

**Exclusion and self-exclusion:** adults only. Participants with a history of significant mental-health difficulties that could be exacerbated by mood induction may self-exclude; the information sheet says so without requiring disclosure. The study does not collect clinical data. If a participant becomes distressed, the experimenter stops the session, stays with them, offers water and a neutral activity, and provides contact details for support services. Adverse events are recorded and reported to the ethics committee.

## 15.4 Deception and debrief

The study uses mild deception: the "buggy search" is intentional, and the purpose is described broadly rather than as an emotion-and-menu study. This is justified only if the deception is necessary (it is: knowing the hypothesis would change behaviour), the risk is minimal, and the debrief is full. The debrief explains the four inductions, the menu conditions, the adaptation hypotheses, and why the search failed. Participants can withdraw their data at that point without penalty. The debrief also checks mood and offers a positive closing task.

## 15.5 Privacy and data protection

| Data | Sensitivity | Handling |
|---|---|---|
| **Consent / contact** | Identifiable | Stored separately from research data; encrypted; access limited to the PI; deleted after the retention period. |
| **Click / timing logs** | Pseudonymous | Participant ID only; no names, emails, IPs, or free-text identifiers; stored on an encrypted server. |
| **Questionnaire responses** | Pseudonymous; affect data | Treated as sensitive; reported in aggregate; free-text quotes checked for identifiers. |
| **Eye tracking (optional)** | Biometric-like; more identifying | Separate consent; store gaze coordinates and AOI metrics, not raw video, unless necessary; separate access; no public sharing of raw gaze. |
| **Audio / video** | High | Not collected in the core study. If interviews are recorded, separate consent and transcription; delete audio after transcription. |

**Legal basis:** for EU/UK participants, consent is the legal basis for processing; the study is not clinical. A data protection impact assessment (DPIA) is completed if eye tracking or large-scale affect data are collected. Data are stored in the EU/UK or in a jurisdiction with an adequacy decision. Retention follows the ethics approval (e.g., five years after publication), after which data are deleted or anonymised. Participants can request deletion of their identifiable data during the retention period; anonymised research data may be exempt from deletion if it is already part of the analysis, as explained in the information sheet.

## 15.6 Compensation and power

Compensate at £12–£15 per hour or course credit with a comparable alternative. Do not make compensation contingent on performance or on completing the frustration induction; that would be coercive. Participants are paid for their time even if they withdraw. The experimenter is not the participant's instructor or supervisor where possible; if they are, a third party handles recruitment and consent to reduce power imbalance.

## 15.7 AI / estimator ethics

The estimator predicts a self-reported state; it does not diagnose a disorder, infer protected characteristics, or make consequential decisions. It runs locally in the experiment and is not deployed to real users. The interface explains when it adapts and why, offers "keep current" and "switch back," and never changes mid-route. The paper reports the estimator's failure modes and does not claim emotional accuracy beyond the tested labels. If the system were deployed commercially, it would need a separate fairness, transparency, and consent review.

## 15.8 Risk register

| Risk | Likelihood | Impact | Mitigation |
|---|---|---|---|
| **Emotional distress** | Low | Medium | Mild inductions, stop rules, debrief, support resources, experimenter training. |
| **Feeling deceived / distrust** | Medium | Medium | Full debrief, explanation of why deception was necessary, right to withdraw data, neutral closing task. |
| **Privacy breach** | Low | High | Pseudonymous IDs, encryption, access control, no third-party scripts, DPIA, retention schedule. |
| **Data loss** | Medium | High | Offline queue, nightly backups, checksum, dry-run network-drop test. |
| **Coercion / power imbalance** | Low | Medium | Third-party recruitment, course-credit alternative, no performance bonus, right to withdraw. |
| **Accessibility exclusion** | Medium | Medium | Keyboard support, contrast, reduced motion, separate keyboard-only session, report input modality. |
| **Researcher bias** | Medium | Medium | Pre-registration, automated scoring, blinded analysis where possible, open code and data. |
| **Adverse event** | Low | High | Stop, support, record, report to ethics committee, follow up. |

# 16 Validity, limitations, and falsification

A strong paper is not one that hides its weaknesses; it is one that anticipates them and designs around them. This section states the threats to inference and the limits of what the study can claim. It also says what evidence would falsify the core hypothesis.

## 16.1 Internal validity

| Threat | Why it matters | Mitigation |
|---|---|---|
| **Menu confounds** | The six menus differ in click depth, visible breadth, persistence, animation, colour, and motor demand. A difference cannot be attributed to "structure" alone. | Parity contract; click-depth mediation; matched-depth sensitivity; colour counterbalancing; common animation duration; process measures. |
| **Emotion induction** | Induced states may be weak, short-lived, or different from real shopping emotions. | Manipulation checks; pilot criterion; re-induction; CACE analysis; dose-response; explicit limitation. |
| **Order and learning** | Participants see six menus four times; they learn the menus and the product taxonomy. | Latin squares; practice; trial-position covariate; first-trial sensitivity; counterbalanced target sets. |
| **Fatigue** | A 90-min session with 24 trials can produce fatigue, especially in the bored condition. | Breaks; neutral reset; pilot timing; fatigue rating; split-session follow-up if needed. |
| **Demand characteristics** | Participants may guess the hypothesis and behave accordingly. | Broad cover story; no emotion labels in the UI; suspicion check; sensitivity analysis excluding guessers; sham control. |
| **Experimenter bias** | The experimenter knows the condition and may unintentionally influence performance. | Standardised script; automated scoring; no performance feedback; second coder for qualitative data; blinded analysis where possible. |
| **Task/target confounds** | Some targets are easier or more semantically transparent than others. | Unique attribute descriptions; difficulty balance; target rotation; target random intercept; pilot calibration. |
| **Carry-over** | A frustrating block may affect the next block; a menu switch may affect the next trial. | Neutral reset; block order counterbalanced; period/sequence covariates; first-trial-after-switch analysis. |

## 16.2 Construct validity

- **Emotion labels:** self-report measures the participant's interpreted state, not a latent "true" emotion. The paper uses "self-reported emotional context" and avoids clinical language.
- **Induction:** a buggy search is a plausible frustration trigger, but it is not the same as real purchase frustration. The ambiguity task is a proxy for category confusion, not for all confusion.
- **Menu constructs:** "structure" is a bundle of geometry, breadth, depth, and persistence. The paper reports the bundle and uses mediation/sensitivity to decompose it; it does not claim to isolate a single dimension.
- **Outcomes:** success and time are objective but task-specific; usability and load are self-reported; engagement is inferred from self-report and behaviour. No single measure is treated as the whole construct.
- **Questionnaires:** SUS, UMUX-Lite, NASA-TLX, and SAM are established, but they were not designed for this exact context; pilot reliability and comprehension checks are required.

## 16.3 External validity

| Dimension | This study | Limit / future work |
|---|---|---|
| **Setting** | Lab, desktop, 1920 × 1080, mouse/trackpad. | Real users browse on mobile, in noisy environments, with different goals. A mobile/field study is future work. |
| **Task** | Known-item product finding with a 120-s cap; search disabled. | Real shopping includes search, browsing, price comparison, and purchase. Study 3 adds free browsing; a field study is needed for purchase behaviour. |
| **Sample** | Adults, mostly university/online panel, English-speaking. | Cultural and age differences in menu familiarity and emotion expression; report demographics and avoid overgeneralising. |
| **Domain** | Fictional watch store, 72 products, four departments. | Other domains have different taxonomy shapes, brand effects, and purchase stakes. Replicate in groceries, electronics, or travel. |
| **Stakes** | No real money, no real purchase. | Real stakes may amplify frustration and change risk preferences; the induced states are mild by design. |
| **Time** | One session, short trials. | Adaptation effects may change with learning over days; a longitudinal study is needed. |

## 16.4 Statistical conclusion validity

- **Power:** the 6 × 4 omnibus is powered for medium effects; small effects will be inconclusive. The paper reports CIs and does not interpret a non-significant interaction as proof of no effect.
- **Multiplicity:** pre-registered families; Holm for focused contrasts; FDR for exploratory cells; all analyses labelled.
- **Model assumptions:** random-effects structure, linearity on the log scale, distributional assumptions; diagnostics and sensitivity models.
- **Missing data:** failures are outcomes, not missing; questionnaire missingness handled with multiple imputation as a sensitivity analysis.
- **Researcher degrees of freedom:** pre-registration, frozen analysis script, open data/code, no optional stopping.
- **Effect sizes:** odds ratios, geometric mean ratios, standardized differences, and precision; practical significance tied to the policy rule.

## 16.5 Falsification: what would prove the core idea wrong?

| Observation | Conclusion |
|---|---|
| **No menu × state interaction, and one menu is best in every state.** | Emotion-contingent structural adaptation is unnecessary in this domain. The paper reports a robust global menu and discusses boundary conditions. |
| **Interaction exists, but matched adaptation is not better than the static best menu.** | The cell-level differences are too small or the switching costs are too high. Adaptation is notworth the disruption in practice. |
| **Matched and mismatched both beat static by the same amount.** | The benefit is change/novelty/attention, not matching. The theoretical claim of affective fit is not supported. |
| **The estimator cannot beat the majority baseline.** | The interaction-signal approach is not viable with these features; the paper reports the negative result and the features that would be needed. |
| **Users systematically revert or report disorientation.** | Stability and control dominate performance; adaptation should be opt-in, not automatic. |

**The strongest version of the paper**

The strongest version does not need every hypothesis to be confirmed. It needs a clean 6 × 4 matrix, a credible induction, a causal test with the right counterfactuals, and an honest estimator evaluation. A paper that says “matching helps for frustration but not for confusion, and the estimator is not yet accurate enough to deploy” is more useful than one that claims a universal win.

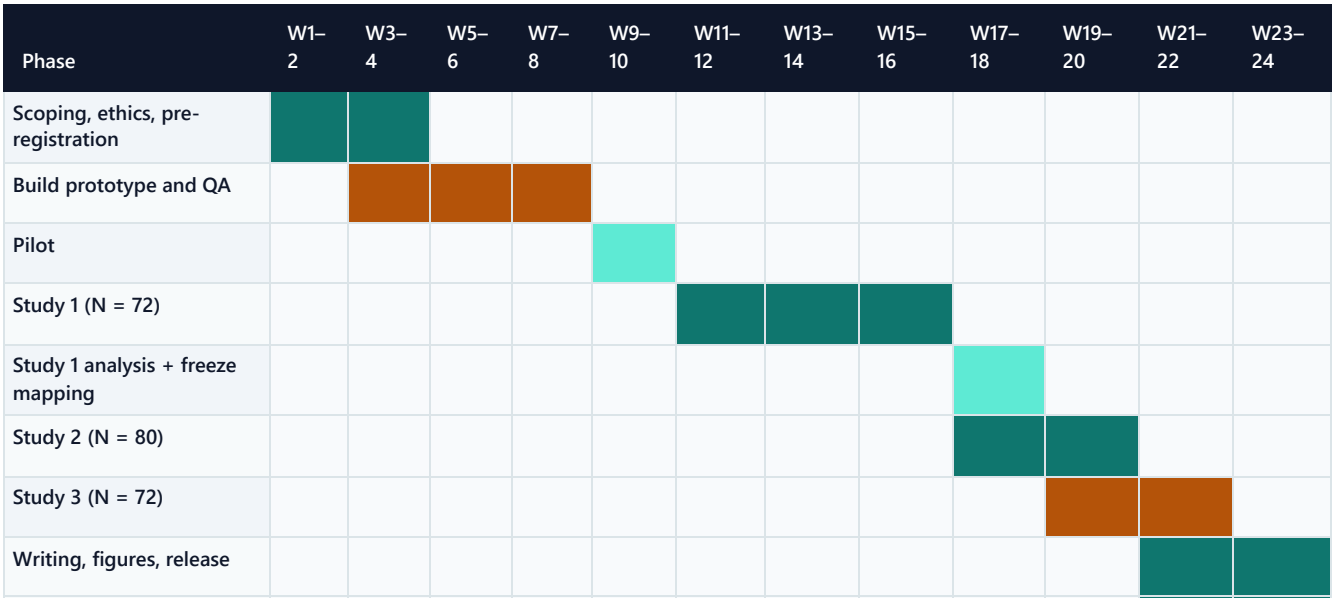
## 16.6 Limitations to state explicitly

1. The induced emotions are mild and temporary; they are not clinical states or necessarily the emotions of real shopping.
2. “Menu structure” is a bundle; the study cannot isolate every visual or motor dimension.
3. The mega menu has a click advantage; the analysis treats it as part of the structure and adds a matched-depth sensitivity analysis.
4. The lab setting and fictional store limit ecological validity; search is disabled in the main task.
5. The sample is not representative; cultural and age differences are not tested.
6. The estimator is trained on a small, lab-generated dataset; its accuracy is an upper bound on what a real deployment could achieve without more data.
7. The study measures short-term adaptation; long-term learning, trust, and habituation are future work.

# 17 Timeline, milestones, and deliverables

The schedule below is for the full programme (Track C). It assumes a part-time researcher with one developer and access to a lab. Track A and Track B compress the same logic into a semester. The critical path is: ethics → build → pilot → Study 1 → freeze mapping → Study 2 → Study 3 → write.

## 17.1 Full programme (24 weeks)




Teal = primary work; amber = build/Study 3; light teal = pilot/analysis. Weeks are indicative; recruitment is the main source of slippage. Build in a 2-week buffer before the pilot and a 2-week buffer before submission.

## 17.2 Milestones and gates

| ID | Milestone | Week | Gate / evidence |
|---|---|---|---|
| M1 | Ethics approval and pre-registration | 2 | Approval letter; OSF preregistration DOI; consent and debrief approved. |
| M2 | Build gate | 8 | All six menus reach all 72 products; parity tests pass; 432-path nightly run; accessibility smoke test. |
| M3 | Pilot sign-off | 10 | Induction criteria met; task difficulty 75–90%; session ≤ 100 min; no blocking bugs. |
| M4 | Study 1 data lock | 16 | N = 72 complete; quality checks; analysis script frozen. |
| M5 | Mapping frozen | 18 | 6 × 4 matrix; policy table; uncertainty flags; Study 2 pre-registration amendment. |
| M6 | Study 2 data lock | 20 | N = 80 complete; condition balance; carry-over checks. |
| M7 | Study 3 data lock | 22 | N = 72 complete; estimator validation; break-even simulation. |
| M8 | Preprint and release | 24 | Manuscript; open materials; data/code DOI; talk/poster. |

## 17.3 Compressed tracks

| Track | Weeks | Schedule | What is cut |
|---|---|---|---|
| A | 8–10 | W1–2 scoping/ethics; W3–5 build; W6 pilot; W7–9 Study 1 (N=48); W10 analysis/write-up. | No Study 2/3; focused contrasts only; descriptive mapping. |
| B | 12–16 | W1–2 scoping/ethics; W3–6 build; W7–8 pilot; W9–12 Study 1 (N=48); W13–14 freeze mapping; W15–18 Study 2 (N=60); W19–20 write-up. | No Study 3; reduced samples; medium effects only. |
| C | 24 | As §17.1. | Nothing, but recruitment must be realistic. |

## 17.4 Deliverables

### Research outputs

- Pre-registration with hypotheses, sample size, and analysis plan.
- 6 × 4 mapping dataset and analysis report.
- Study 2 causal report and policy recommendation.
- Study 3 estimator validation and break-even curve.
- Manuscript (CHI/IUI/TOCHI style) and preprint.
- Conference talk or poster; thesis chapter.

### Engineering outputs

- Open-source six-menu prototype with a shared taxonomy.
- Event schema and analysis pipeline.
- Task bank and product dataset (fictional).
- Reproducibility package (Docker, seeds, README).
- Accessibility and parity test suite.
- Data dictionary and codebook.

# 18 Paper outline and publication plan

The paper should be written as an empirical HCI paper with a clear causal claim and a honest estimator evaluation — not as a design showcase. The title, abstract, and contribution statement should foreground the 6 × 4 mapping and the matched-vs-static test.

## 18.1 Title and abstract

**Working title:** "Emotion-Contingent Navigation: A Controlled Comparison of Six Menu Structures and a Causal Test of Adaptive Switching."

**Alternative:** "Does the Menu Need to Change? Mapping and Testing Emotion-Contingent Navigation in E-Commerce."

**Abstract structure (200–250 words):** (1) problem — static menus, emotional context, adaptive interfaces; (2) gap — no controlled test of whole-structure adaptation; (3) method — pilot, 6 × 4 within-subject mapping, causal test with static/matched/mismatched/sham, end-to-end estimator; (4) results — the key pattern (to be filled); (5) implication — when adaptation helps, when it does not, and the accuracy threshold.

## 18.2 Section plan

| Section | Words | Content / figures |
|---|---|---|
| **1. Introduction** | 1,200 | Problem, motivating example, gap, contributions, roadmap. Figure 1: the six menus. |
| **2. Related work** | 2,000 | Menu structure; adaptive menus and switching costs; emotion in HCI; navigation/lostness; the four supplied papers; theoretical framework. |
| **3. Study 1: Mapping** | 3,000 | Design, participants, materials, procedure, measures, analysis, results (6 × 4 matrix), manipulation check, discussion. Figures 2–3; Tables 2–3. |
| **4. Study 2: Causal test** | 2,500 | Conditions, procedure, results (matched vs static/mismatched/sham), switching cost, mediation, discussion. Figure 4; Tables 4–5. |
| **5. Study 3: End-to-end** | 2,500 | Estimator, validation, hybrid vs static, confidence stratification, break-even simulation, discussion. Figures 5–6; Tables 6–7. |
| **6. General discussion** | 2,000 | When adaptation helps; mechanism; design implications; the cost of switching; the accuracy threshold; comparison with RL/adaptive-menu work. |
| **7. Limitations** | 800 | Induction, menu bundle, lab, sample, short-term, estimator. |
| **8. Conclusion** | 400 | Three take-aways: map first, test causally, abstain when uncertain. |

**Core figures:** (1) six menu wireframes; (2) 6 × 4 matrix heatmap of success/time; (3) interaction/posterior plot; (4) matched vs static vs mismatched vs sham; (5) estimator confusion matrix and calibration; (6) break-even utility curve. **Core tables:** menu operationalisation; participant demographics; manipulation check; primary mixed-model results; condition contrasts; estimator performance.

## 18.3 Contribution statement (draft)

1. We provide the first controlled 6 × 4 mapping of menu-structure performance and experience across experimentally induced emotional contexts in e-commerce.
2. We separate *matching* from *mere change* using static, matched, mismatched, and sham conditions, and we quantify switching costs.
3. We show how to derive a policy from the mapping with explicit uncertainty, and we test it end-to-end with an imperfect estimator.

4. We report the estimator accuracy at which adaptation stops paying for itself, giving designers a concrete threshold rather than a slogan.
5. We release a reproducible six-menu prototype, task bank, event schema, and analysis pipeline.

## 18.4 Target venues and fit

| Venue | Fit | What to emphasise |
|---|---|---|
| **ACM CHI** | Strong fit for the mapping + causal test; full paper. | Novel interaction comparison; rigorous design; open materials; design implications. |
| **ACM IUI** | Strong fit for the adaptive policy and estimator. | Personalisation, uncertainty, user control, break-even accuracy. |
| **ACM CSCW** | Moderate fit if the framing emphasises everyday shopping and user agency. | User control, trust, opt-in adaptation, real-world context. |
| **ACM TOCHI / IJHCS** | Strong fit for a journal-length version with all three studies. | Theory, mechanism, generalisation, full method detail. |
| **Electronic Commerce Research and Applications** | Good fit for the e-commerce framing. | Practical implications, conversion, menu design, consumer affect. |
| **CHI EA / local HCI conference** | Good for an early work-in-progress or thesis preview. | Mapping design, pilot results, demo. |

## 18.5 Reviewer objections and planned responses

| Likely objection | Response / design feature |
|---|---|
| **"The menus differ in click depth; the comparison is unfair."** | Click depth is a real property of mega menus. Report it, mediate it, and add a matched-depth sensitivity analysis on A/B/D/E/F. |
| **"Induced emotions are not real shopping emotions."** | Acknowledge; use manipulation checks, dose-response, and a clear scope statement. A field study is future work. |
| **"Six menus × four states is underpowered."** | Report the simulation, the achieved precision, and the focused contrasts; do not overclaim small effects. Use N=72 where possible. |
| **"The adaptive gain could be novelty."** | The mismatched and sham conditions directly test this; report the pattern honestly. |
| **"The estimator is trained and tested on lab data."** | Participant-grouped CV, frozen pipeline, calibration, break-even curve; discuss deployment gap. |
| **"Why these four emotions?"** | Valence × arousal + cognitive/motivational distinctness; alternatives considered; manipulation checks and discriminant validity. |
| **"Why these six menus?"** | Coverage of linear, panel, recursive, icon-grid, and radial geometries; each has a theoretical rationale; the set is not exhaustive. |
| **"Where is the theory?"** | Cognitive load, wayfinding/spatial memory, visual salience, motor efficiency; mediation tests; predictions in Table 1.2. |

## 18.6 Publication strategy

**Paper 1 (core):** Study 1 + Study 2, with the estimator break-even simulation as a bridge. This is a complete CHI/IUI paper: mapping, causal test, and a practical threshold. **Paper 2 (extension):** Study 3 end-to-end with the full estimator, calibration, and field deployment. If time is short, publish Paper 1 and treat Study 3 as a thesis chapter or

## 18.7 Open science

- Pre-register on OSF before data collection; update with deviations and their reasons.
- Release the prototype, taxonomy, task bank, event schema, and analysis code under a permissive licence.
- Share anonymised trial-level data and questionnaires if ethics allow; otherwise share a synthetic dataset and the code.
- Report all conditions, all pre-registered outcomes, and all exclusions; do not hide null results.
- Archive a DOI (Zenodo) for the exact manuscript version.

# 19 Conclusion

The project's contribution is not "emotion-aware menus" as a slogan. It is a disciplined answer to three questions: **which menu structure fits which emotional context**, **whether using that mapping causally helps**, and **how accurate the emotion estimate must be before adaptation is worth it**.

The plan is deliberately conservative. It treats emotion as a self-reported context, not a hidden truth. It treats menu structure as a bundle of geometry, breadth, depth, and persistence, and it measures the confounds rather than pretending they do not exist. It uses a static best menu, a mismatched menu, and a sham switch as counterfactuals, because "adaptive beats static" is not enough. It tests the estimator separately from the policy, because a good mapping with a bad estimator can still fail. And it pre-registers what would count as failure.

If the result is positive, the paper offers a validated mapping, a causal estimate, and a deployable policy with a confidence threshold. If the result is null, the paper still offers something useful: evidence that a stable menu is robust to mild emotional context, or that switching costs dominate, or that the estimator is the bottleneck. Either way, the study replaces a plausible intuition with evidence.

**Map**
6 menus × 4 states, with uncertainty

**Test**
matched vs static vs mismatched vs sham

**Estimate**
interaction signals → state, with calibration

**Decide**
when to adapt, when to abstain

**Immediate next steps:** confirm the timeline and sample access; obtain ethics approval; build the taxonomy and task bank; prototype the six menus; run the pilot; freeze the protocol; pre-register. The rest of the plan follows from those five steps.

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If either PDF is obtained, verify the exact title, author list, DOI, and reported findings before adding it to the reference list. If not, remove the attribution and describe the menu as a design pattern.

## 20.1 Source-to-menu map

| Menu | Primary source | What the source actually supports |
|---|---|---|
| **A · Horizontal** | Chiu & Yang (2024); Leuthold et al. (2011) | Eye-tracking comparisons of horizontal vs vertical/dynamic menu formats; visual attention and preference. Not a test of emotional context. |
| **B · Sidebar** | Chiu & Yang (2024); Leuthold et al. (2011) | Vertical menu format, persistence, and scanning behaviour. Not a test of induced emotion. |
| **C · Mega** | Li & Nielsen (2017); Protocol v3.0 | Practical guidance on mega menus and the protocol's baseline design. No controlled emotion study. |
| **D · Fractal** | Sahraoui et al. (2025) | Concept and engineering of fractal adaptive menus; use case, not a controlled comparison. |
| **E · RS grid** | UNVERIFIED Chiossi et al. (2026); Ahlström et al. (2010) | Only the general square/icon menu literature is verified. The specific RS design is operationalised in this plan, not attributed. |
| **F · Radial** | Kim et al. (2022); UNVERIFIED Collins et al. (2026) | Lattice menu with visual anchors for gaze; adapted from gaze to mouse. The Collins citation is unverified. |

# A Appendices

The appendices give the concrete materials a researcher needs to run the study: the task bank, questionnaire items, induction scripts, data schema, randomisation matrices, analysis checklist, pilot QA checklist, consent/debrief outline, and glossary. The full machine-readable task bank and prototype are part of the repository, not this PDF.

## Appendix A — Task bank, difficulty rubric, and 24-task sample

### A.1 Generation rules

1. Every leaf product receives a unique attribute vector: case size, strap, movement, water resistance, dial colour, and price band.
2. A task description uses **two or three** attributes and never the exact leaf label. It must map to exactly one product; the build-time uniqueness test rejects ambiguous descriptions.
3. Tasks are written in natural shopping language ("You're looking for…"), not menu language ("Go to Sport & Dive").
4. Each task is assigned a difficulty level by the rubric below; the final bank is balanced 24 easy / 24 medium / 24 hard.
5. Each participant sees 24 tasks, balanced 8 / 8 / 8 across difficulty and across departments; the target set is rotated across participants.

### A.2 Difficulty rubric

| Level | Position in the menu | Semantic distance | Distractors |
|---|---|---|---|
| **Easy** | Department 1–2; subcategory 1; leaf 1–2. | Task words overlap strongly with the leaf label. | 0–1 same-colour or same-material leaves. |
| **Medium** | Department 2–3; subcategory 2; leaf 3–4. | One attribute must be translated (e.g., "lightweight" → titanium). | 2–3 plausible leaves in the same subcategory. |
| **Hard** | Department 3–4; subcategory 3; leaf 5–6. | Two attributes are indirect; the label uses a metaphor or material term. | 4+ plausible leaves across subcategories. |

### A.3 24-task sample

Table A.1 — Illustrative task sample. The repository contains the full 72-task bank and the uniqueness test output.

| ID | Task wording | Department | Subcategory | Leaf | Diff. |
|---|---|---|---|---|---|
| **T01** | Titanium dive watch, 300 m, deep blue dial | Sport & Dive | Pro Diver 300m | Steel Blue | Easy |
| **T02** | Lightweight titanium dive watch, 300 m, black dial | Sport & Dive | Pro Diver 300m | Titanium Deep | Medium |
| **T03** | 42 mm automatic trail watch, green dial, fabric strap | Sport & Dive | Trail Master GMT | Forest Green | Medium |
| **T04** | Rose-gold dress watch, ivory dial, leather strap | Dress & Formal | Heritage Classic | Ivory Dial | Easy |
| **T05** | Black-tie watch, deep navy dial, 40 mm | Dress & Formal | Midnight Dress | Deep Navy | Medium |
| **T06** | Moonphase complication, celestial blue dial | Dress & Formal | Artisan Moonphase | Celestial Blue | Hard |
| **T07** | 38 mm field watch, khaki canvas strap, automatic | Everyday & Casual | Field Standard | Khaki Canvas | Easy |
| **T08** | Minimalist watch, matte white dial, steel mesh strap | Everyday & Casual | Urban Minimal | Matte White | Medium |
| **T09** | Weekend watch, sunset clay dial, rubber strap | Everyday & Casual | Weekender | Sunset Clay | Hard |
| **T10** | Graphite smartwatch, titanium case, 44 mm | Smart & Hybrid | Smart One | Graphite | Medium |
| **T11** | Hybrid sport watch, orange accent, rubber strap | Smart & Hybrid | Hybrid Active | Orange Sport | Hard |
| **T12** | Connected classic, burgundy leather strap, 42 mm | Smart & Hybrid | Connected Classic | Burgundy | Hard |
| **T13** | Steel dive watch, 300 m, ocean-green dial | Sport & Dive | Pro Diver 300m | Ocean Green | Medium |
| **T14** | Chronograph, white dial, steel bracelet | Sport & Dive | Aqua Racer Chrono | Regatta White | Easy |
| **T15** | GMT trail watch, summit-blue dial, steel strap | Sport & Dive | Trail Master GMT | Summit Blue | Medium |

| | | | | | |
|---|---|---|---|---|---|
| **T16** | Classic dress watch, anthracite dial, steel case | Dress & Formal | Heritage Classic | Anthracite | Medium |
| **T17** | Moonphase, warm-silver dial, leather strap | Dress & Formal | Artisan Moonphase | Warm Silver | Hard |
| **T18** | Field watch, olive dial, fabric strap | Everyday & Casual | Field Standard | Olive | Easy |
| **T19** | Minimalist watch, concrete-grey dial | Everyday & Casual | Urban Minimal | Concrete Grey | Medium |
| **T20** | Weekend watch, cream dial, leather strap | Everyday & Casual | Weekender | Cream | Hard |
| **T21** | Hybrid sport watch, silver case, blue sport strap | Smart & Hybrid | Hybrid Active | Blue Sport | Medium |
| **T22** | Connected classic, cream dial, brown leather strap | Smart & Hybrid | Connected Classic | Cream | Hard |
| **T23** | Dive watch, bronze case, green dial | Sport & Dive | Pro Diver 300m | Bronze Reef | Hard |
| **T24** | Dress watch, champagne dial, black leather strap | Dress & Formal | Midnight Dress | Champagne | Hard |

Some descriptions above are intentionally close to a label; the build-time test, not the table, is the authority. If two products match, the task is rewritten or the product attribute vector is changed before the pilot.

# Appendix B — Questionnaire items

## B.1 Self-Assessment Manikin (SAM)

**Valence:** 9-point pictorial scale, 1 = very negative, 9 = very positive. **Arousal:** 9-point pictorial scale, 1 = very calm, 9 = very aroused. Anchor: "How do you feel *right now*?"

## B.2 Discrete emotion check

"Right now, how much do you feel each of the following?" 7-point scale (1 = not at all, 7 = extremely): **happy**, **frustrated**, **confused**, **bored**. Add: "None of these describes how I feel" (checkbox). Dominant-state rule: the highest item must be ≥ 4 and at least 1 point above the second-highest; otherwise the state is labelled *mixed*.

## B.3 NASA-TLX workload (frustration subscale excluded)

Four items, 0–100: **mental demand** ("How mentally demanding was the task?"), **temporal demand** ("How hurried or rushed was the pace?"), **effort** ("How hard did you have to work to accomplish your level of performance?"), **performance** ("How successful were you in accomplishing what you were asked to do?"; reverse-scored). Report the mean or weighted score; report the standard frustration item separately, never inside the workload composite.

## B.4 UMUX-Lite

1 = strongly disagree, 7 = strongly agree. (1) "The interface's capabilities meet my requirements." (2) "The interface is easy to use." Score: `((item1 − 1) + (item2 − 1)) / 12 × 100`.

## B.5 System Usability Scale (SUS)

10 items, 1 = strongly disagree, 5 = strongly agree. Standard alternating positive/negative wording (Brooke, 1996). Score: odd items − 1; 5 − even items; sum × 2.5. Administer once per menu at the end of the session, with menu order counterbalanced.

## B.6 Disorientation / control

7-point agreement: (1) "I felt lost." (2) "I knew where I was in the store." (3) "I could predict where to go next." (4) "I felt in control of the navigation." Items 2–4 reverse-scored for a disorientation score; also report the four items separately.

## B.7 Engagement (short form)

7-point agreement: (1) "I was absorbed in the task." (2) "The interface was aesthetically appealing." (3) "The task felt worthwhile." (4) "I wanted to continue exploring." Adapted from O'Brien et al. (2018); report as a short engagement index, not as the full UES.

## B.8 Trust / perceived intelligence (Study 2/3)

7-point agreement: (1) "The interface understood my needs." (2) "The change was appropriate for the task." (3) "I trusted the system." (4) "The interface felt intelligent."

## B.9 Adaptation experience (Study 2/3)

(1) "I noticed that the interface changed." (2) "I understood why it changed." (3) "The change disrupted my flow." (4) "I would want this adaptation in a real store." (5) "I would prefer the previous layout." Plus behavioural counts: keep-current clicks, switch-back clicks, dislike

clicks.

### B.10 Demographics and individual differences

Age; gender (open field); education; online shopping frequency (never / monthly / weekly / daily); watch expertise (1–7); tech savvy (1–7); 4-item need for cognition; 4-item ambiguity tolerance; colour-vision screening; prior experience with each menu type (1–7).

### B.11 Suspicion check

Open-ended: "What did you think this study was about?" "Did you notice any pattern in the menus or the way they changed?" "Did you change your behaviour because you thought the study wanted you to?" Code as *no hypothesis*, *partial*, *correct guess*.

## Appendix C — Induction scripts

**Happy (≈ 2.5 min).** "For the next two minutes, please watch this short clip. It's a light comedy scene; there's nothing you need to do except watch. You can look away or stop if you prefer." *[Play the clip.]* "Thanks. Take a moment. How do you feel right now?" *[SAM + discrete check.]*

**Frustration (≈ 2.5 min).** "Please use the search box to find the 'Titanium Deep' watch. The search is still being tested, so it may be slow." *[The search fails with a generic "No results — try again" message.]* "Please try once more." *[Second failure.]* "One more try, please." *[Third failure.]* "Thank you — that's enough for this part. The search was designed to fail; it wasn't your fault." *[Record frustration; offer a 30-s neutral pause before the trials.]*

**Confusion (≈ 2.5 min).** "Here are six watches. Please sort them into the two categories you think the store uses. There's no right answer — we're interested in how people make sense of ambiguous categories. Talk me through your thinking as you go." *[The items are deliberately ambiguous: two could fit either category; no feedback is given.]* "Thanks. That was intentionally ambiguous; the ambiguity is the point of this part."

**Boredom (≈ 90 s).** "Please check the price of each watch on this list and write it on the sheet. It's a routine data-entry task; accuracy matters more than speed." *[20 near-identical rows; no feedback; repeat until 90 s.]* "Thanks — that's the end of the data-entry task."

**Neutral reset (≈ 60 s).** "Let's take a short pause. Please watch the screen; there's nothing to remember or do." *[Play 60 s of calm nature footage, no abrupt cuts or loud audio.]* "When you're ready, we'll continue."

**Stop rules**

Stop the induction immediately if the participant says "stop," rates distress > 6/10, shows signs of panic or tearfulness, or asks to leave. Stay with them, offer water and a neutral activity, and debrief. Record the event and report it to the ethics committee. Never continue a block "to complete the data."

# Appendix D — Data schema (SQL DDL, simplified)

The DDL is split into five blocks so that it paginates cleanly. It is illustrative; the migration files in the repository are the source of truth.

## D.1 Participants and sessions

```
CREATE TABLE participants (
  participant_id   text PRIMARY KEY,
  created_at       timestamptz NOT NULL DEFAULT now(),
  age_band         text,
  gender           text,
  education        text,
  shopping_freq    text,
  watch_expertise  smallint,
  tech_savvy       smallint,
  colour_vision    text,
  consent_version  text NOT NULL
);

CREATE TABLE sessions (
  session_id       uuid PRIMARY KEY,
  participant_id   text REFERENCES participants,
  study            text NOT NULL,          -- pilot | s1 | s2 | s3
  sequence_id      text NOT NULL,
  taxonomy_hash    text NOT NULL,
  started_at       timestamptz NOT NULL,
  ended_at         timestamptz,
  input_type       text,
  browser          text,
  viewport         text
);
```

## D.2 Trials

```
CREATE TABLE trials (
  trial_id         uuid PRIMARY KEY,
  session_id       uuid REFERENCES sessions,
  block_id         text NOT NULL,
  trial_index      smallint NOT NULL,
  task_id          text NOT NULL,
  target_id        text NOT NULL,
  menu_id          text NOT NULL,
  state_label      text NOT NULL,
  condition        text,
  success          boolean,
  time_ms          integer,
  clicks           smallint,
  errors           smallint,
  backtracks       smallint,
  category_switches smallint,
  path_efficiency  numeric(5,3),
  first_click_ok   boolean,
  search_attempts  smallint,
  timed_out        boolean,
  gave_up          boolean
);
```

### D.3 Events

```
CREATE TABLE events (
  event_id         uuid PRIMARY KEY,
  session_id       uuid REFERENCES sessions,
  trial_id         uuid REFERENCES trials,
  event_type       text NOT NULL,
  t_client         double precision NOT NULL,
  t_server         timestamptz NOT NULL DEFAULT now(),
  elapsed_ms       integer,
  level            smallint,
  element_id       text,
  element_label    text,
  parent_id        text,
  x                integer,
  y                integer,
  meta             jsonb NOT NULL DEFAULT '{}'::jsonb
);
```

### D.4 Questionnaires and randomisation audit

```
CREATE TABLE questionnaires (
  response_id      uuid PRIMARY KEY,
  session_id       uuid REFERENCES sessions,
  block_id         text,
  menu_id          text,
  instrument       text NOT NULL,
  item_id          text NOT NULL,
  value            numeric,
  text_value       text,
  answered_at      timestamptz NOT NULL DEFAULT now()
);

CREATE TABLE randomisation_audit (
  participant_id   text PRIMARY KEY,
  sequence_id      text NOT NULL,
  block_order      text NOT NULL,
  menu_orders      jsonb NOT NULL,
  target_set       smallint NOT NULL,
  seed_hash        text NOT NULL,
  assigned_at      timestamptz NOT NULL DEFAULT now()
);
```

### D.5 Adaptation log and indexes

```
CREATE TABLE adaptation_log (
  decision_id      uuid PRIMARY KEY,
  session_id       uuid REFERENCES sessions,
  trial_id         uuid REFERENCES trials,
  predicted_state  text,
  confidence       numeric(4,3),
  current_menu     text,
  target_menu      text,
  decision         text NOT NULL,   -- adapt | abstain | rejected | reverted
  reason           text,
  created_at       timestamptz NOT NULL DEFAULT now()
);

CREATE INDEX idx_events_session ON events(session_id, t_client);
CREATE INDEX idx_trials_session ON trials(session_id, block_id);
```

# Appendix E — Randomisation matrices

## E.1 Emotion block order (4 × 4 Latin square)

| Sequence | Position 1 | Position 2 | Position 3 | Position 4 |
|---|---|---|---|---|
| **S1** | Happy | Frustrated | Confused | Bored |
| **S2** | Frustrated | Confused | Bored | Happy |
| **S3** | Confused | Bored | Happy | Frustrated |
| **S4** | Bored | Happy | Frustrated | Confused |

## E.2 Menu order within a block (6 × 6 Latin square)

| Row | Pos 1 | Pos 2 | Pos 3 | Pos 4 | Pos 5 | Pos 6 |
|---|---|---|---|---|---|---|
| **R1** | A | B | C | D | E | F |
| **R2** | B | C | D | E | F | A |
| **R3** | C | D | E | F | A | B |
| **R4** | D | E | F | A | B | C |
| **R5** | E | F | A | B | C | D |
| **R6** | F | A | B | C | D | E |

## E.3 Target-set rotation

Six target sets (T1–T6), each with 24 tasks balanced by department and difficulty. The assignment matrix below is repeated twice to reach N = 72; each row is used once per cycle. The server assigns the next unused row within the stratum, so each target set appears equally often with each block order and menu order.

| Block order | Set for S1 | Set for S2 | Set for S3 | Set for S4 | Set for S5 | Set for S6 |
|---|---|---|---|---|---|---|
| **S1** | T1 | T2 | T3 | T4 | T5 | T6 |
| **S2** | T2 | T3 | T4 | T5 | T6 | T1 |
| **S3** | T3 | T4 | T5 | T6 | T1 | T2 |
| **S4** | T4 | T5 | T6 | T1 | T2 | T3 |
| **S5** | T5 | T6 | T1 | T2 | T3 | T4 |
| **S6** | T6 | T1 | T2 | T3 | T4 | T5 |

The table is illustrative: the actual generator uses a seeded permutation within these constraints and writes an audit row for every assignment. Do not hand-assign conditions.

# Appendix F — Analysis checklist

| Stage | Check |
|---|---|
| **Before data** | Pre-registration posted; primary outcomes fixed; sample size justified; analysis script frozen; random seeds recorded; data dictionary approved. |
| **Data lock** | Export raw JSONL and CSV; checksum; missingness report; exclusion log; randomisation balance; manipulation-check summary. |
| **Descriptives** | Demographics; baseline affect; prior menu experience; input modality; session length; technical errors. |
| **Primary model** | Success GLMM; time LMM; interaction test; estimated marginal means; 95% CIs; effect sizes. |

| | |
|---|---|
| | omnibus gate fails. |
| **Diagnostics** | Residual plots; overdispersion; singularity; collinearity; influence; convergence warnings; sensitivity to random-effects structure. |
| **Sensitivity** | A/B/D/E/F only (exclude C); per-protocol state labels; exclude first trial per block; exclude correct hypothesis guessers; input modality covariate; Bayesian model. |
| **Study 2** | Matched vs static, matched vs mismatched, matched vs sham; non-inferiority/equivalence; carry-over; concordant states excluded; mediation. |
| **Study 3** | Participant-grouped CV; macro-F1; balanced accuracy; confusion matrix; calibration; confidence stratification; switch cost; break-even curve. |
| **Reporting** | Flow diagram; all pre-registered outcomes; all exclusions; effect sizes with CIs; exact p; data/code availability; deviations from pre-registration. |

# Appendix G — Pilot QA checklist

- All 72 products reachable in all six menus; no dead ends; no overlapping hit targets below 44 × 44 px.
- Every task description maps to exactly one product; the uniqueness test passes for the full bank.
- Event log reconstructs every practice and main trial; no missing `trial_end`; no duplicate `event_id`.
- Randomisation balance: each block order, menu position, and target set used approximately equally.
- Questionnaire flow: no missing items; skip logic correct; timing acceptable; mobile/desktop rendering checked.
- Induction: dominant-state rate ≥ 60%; mean intensity ≥ 5/7; no distress > 6/10; stop rules understood by the experimenter.
- Task difficulty: overall success 75–90%; easy > medium > hard; no floor or ceiling.
- Timing: session ≤ 100 min; trial median ≤ 45 s; frame rate ≥ 50 fps; no long tasks during a trial.
- Accessibility: keyboard path works; focus visible; contrast passes; reduced motion respected; axe-core clean.
- Data: offline queue survives a network drop; backup restores; export checksum matches.
- Debrief: deception explained; mood checked; support resources provided; withdrawal option clear.
- Freeze: build hash, taxonomy hash, questionnaire version, and analysis script version recorded.

# Appendix H — Consent and debrief outline

## H.1 Information sheet (plain language)

- **Purpose:** compare website navigation layouts for finding products.
- **What you will do:** product-finding tasks, short mood exercises (a comedy clip, a search task that may not work, an ambiguous sorting task, a repetitive catalogue task), questionnaires.
- **Time:** about 90 minutes (Study 1), 65 minutes (Study 2), 70 minutes (Study 3).
- **Data:** clicks, timing, questionnaire responses; optional eye tracking in a separate session.
- **Risks:** mild, temporary frustration, confusion, or boredom; you can stop at any time.
- **Compensation:** £12–£15/hour or course credit with an alternative; paid even if you withdraw.
- **Privacy:** pseudonymous ID; no names or emails in the research data; retention per ethics approval.
- **Contact:** PI name, email, ethics committee contact, support resources.

## H.2 Debrief script

> "Thank you for taking part. I can now explain the full purpose. We are comparing six different menu layouts and asking whether the best layout depends on how someone feels. The search task earlier was designed to fail — that was part of the study, not your mistake. The sorting task was deliberately ambiguous, and the catalogue task was deliberately repetitive. We used these short exercises to create mild, temporary states; they are not a test of you. The interface sometimes changed between tasks; we were testing whether that change helped or disrupted you. You can withdraw your data now without giving a reason. Do you have any questions? How are you feeling now?" *[Offer a neutral-positive closing task and support resources.]*

# Appendix I — Glossary

| Term | Definition in this plan |
|---|---|
| **Adaptive navigation** | A system that changes the menu structure, content, or presentation in response to the user's state or behaviour. |
| **Affective fit** | The hypothesis that a menu's structural demands should match the processing style associated with the current emotional context. |
| **Breadth** | The number of options visible at a given menu level. |
| **Depth** | The number of levels or clicks required to reach a leaf. |
| **Fractal menu** | A recursive layout in which the same spatial pattern repeats at each level; selecting a parent zooms into its children. |
| **Mega menu** | A wide panel that shows multiple subcategories and leaves at once, usually two clicks from the top level. |
| **Radial lattice** | A menu arranged around a centre, with options at compass sectors and concentric rings. |
| **RS grid** | A grid of large rounded-square icon cards, colour-coded by category. |
| **State label** | The dominant self-reported emotion after induction, or *mixed* if no item dominates. |
| **Matched condition** | The menu selected by the Study 1 policy for the current state. |
| **Mismatched condition** | A pre-registered derangement of the state→menu mapping; controls for change without fit. |
| **Sham condition** | A banner/announcement of change with no actual layout change; controls for demand and attention. |
| **Break-even accuracy** | The estimator accuracy at which the expected utility of hybrid adaptation equals that of the static best menu. |